

Causal Embedding Models for Procedural Documentation: A Survey of Methods for Follow-up Reranking and Workflow Ordering

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Abstract

We study the design of *causal embedding models*: representations and scoring functions that capture the *directional*, *precedence* structure of procedural documentation rather than mere topical similarity. Our concrete setting is the documentation and video tutorials of Adobe Experience Platform (AEP) and Adobe Journey Optimizer (AJO), from which a model must (i) judge whether one step *causally precedes* / *enables* another, (ii) **rerank candidate follow-up questions** by logical progression, and (iii) **sort multi-step workflows into ascending order**. We survey the landscape end to end—causal representation learning and causal reasoning from text; the state of the art in dense retrieval and bi-/cross-/late-interaction architectures; *asymmetric and order-structured embedding geometries* (order, box, Gaussian, and hyperbolic/Lorentz embeddings) that natively express directionality; reasoning rerankers and reinforcement learning for ranking in sub-1B models; sentence/step ordering and robust rank aggregation; and follow-up/conversational retrieval and procedural-workflow understanding. We ground the survey in an empirical characterization of the project’s own data and baselines—where a bi-encoder wins pairwise classification but a DeBERTa cross-encoder wins workflow ordering (21/30 vs. 0/30), exposing that *pairwise accuracy* $\neq$ *ordering quality*. We argue that the dominant failure mode is not the pairwise model but the *aggregation* (naive topological sort collapses on a single cycle) and the *data* (adjacent-only, total-order labels). We synthesize a concrete, sub-1B system design—calibrated cross-encoder $\rightarrow$ minimum-feedback-arc-set / Kemeny aggregation $\rightarrow$ partial-order evaluation, with distilled “think-free” causal rationales and an optional ordering-reward GRPO polish—and release a reference implementation of the aggregation, calibration, metrics, and reranking components.

1 Introduction

1.1 Motivation

Enterprise documentation is overwhelmingly *procedural*: it tells a user how to accomplish a goal through an ordered sequence of steps. Adobe Experience Platform (AEP) and Adobe Journey Optimizer (AJO) ship thousands of documentation pages and video tutorials describing how to build audiences, enrich profiles, design journeys, and orchestrate campaigns. A retrieval or assistant system over this corpus must do more than return topically similar passages—it must understand *what comes before what*. Concretely, three downstream capabilities motivate this survey:

1. **Directional causal judgment.** Given two pieces of text (steps, phases, or doc fragments) t_1 and t_2 , estimate $P(t_1 \text{ causally precedes / enables } t_2)$.
2. **Follow-up question reranking.** Given a user question, order a slate of candidate follow-up questions by logical/causal progression, not just relevance.
3. **Workflow ordering.** Given the unordered steps of a workflow, sort them into the correct ascending execution order.

All three rest on a single primitive—a *directional*, asymmetric notion of precedence—that standard semantic embeddings, built around symmetric cosine similarity, do not natively provide. We call a model that captures this primitive a *causal embedding model*, and we use the term deliberately broadly: it spans asymmetric scoring heads over encoders, order/box/hyperbolic geometries, cross-encoder reasoners, and small generative rerankers. The foundations section clarifies precisely which rung of Pearl’s causal ladder this task occupies (largely structural precedence, not intervention).

1.2 Why this is hard

Three difficulties recur throughout the survey. First, *symmetry*: the workhorse similarity function $\cos(\mathbf{a}, \mathbf{b})$ is symmetric, so a vanilla embedding cannot, by construction,

distinguish “ A enables B ” from “ B enables A .” Second, *causal direction is genuinely hard*: even frontier LLMs perform poorly on pure causal-direction benchmarks, and supervised systems cap well below ceiling. Third, and most consequentially for us, *local correctness does not compose into global correctness*: a model can be accurate on most individual pairs yet still produce a wholly wrong global order, because the step that turns $O(n^2)$ noisy pairwise judgments into a single ranking is fragile. In this project, a DeBERTa-v3 cross-encoder that orders a 6-step workflow nearly correctly and scores 21/30 on pairwise direction nonetheless yields **0/30** correct *full orderings* under a naive topological sort—a gap this survey takes as its central diagnostic.

1.3 Scope and contributions

This document is both a *survey* and a *system-design study* grounded in a real, ongoing project under a strict **sub-1B-parameter** budget and a 4×A100 environment. Our contributions are:

- A structured survey of the methods relevant to causal/directional embeddings, spanning eight areas: causal foundations, embedding/retrieval architectures, causal reasoning from text, asymmetric/order-structured geometry, reasoning rerankers and RL, sentence-ordering and rank aggregation, follow-up/conversational retrieval, and procedural-workflow understanding.
- An empirical characterization of the project’s corpus, pairwise datasets, tier system, and follow-up data, with the baselines any new model must beat.
- A diagnosis that reframes the workflow-ordering failure as primarily an *aggregation* and *data* problem rather than a pairwise-model problem, and a concrete sub-1B system design that follows from it.
- A reference implementation (released alongside this survey) of robust ordering (minimum-feedback-arc-set, Bradley–Terry, Kemeny), score calibration, partial-order metrics, and a model-free follow-up reranker.

1.4 Organization

Section 2 establishes causal foundations and a working definition of “causal embedding.” Section 3 surveys dense-retrieval architectures and modern embeddings. Section 4 covers causal reasoning and direction from text. Section 5 develops asymmetric and order-structured geometries—the most direct route to directionality. Section 6 surveys reasoning rerankers and RL for ranking under a small-model budget; Section 7 covers turning pairwise scores into a global order. Section 8 addresses follow-up and conversational retrieval; Section 9, procedural-workflow understanding. Section 10 characterizes the data. Section 11 synthesizes a recommended system and roadmap, and Section 12 concludes.

2 Foundations: What Is a “Causal” Embedding?

The phrase “causal embedding” is used loosely across the retrieval and representation-learning literature, often as a synonym for any embedding whose geometry encodes *asymmetric* relations. This section pins down what we mean precisely, separates three distinct notions that are routinely conflated, and locates the project’s goal — directional precedence between documentation steps — on the firmest defensible footing. We are deliberately conservative: we argue that a “causal embedding” for procedural documentation is, in the honest analysis, a model of *structural / directional precedence learned from text*, sitting almost entirely at the lowest rung of Pearl’s ladder, and we say so plainly rather than overclaiming interventional or counterfactual content.

2.1 Three notions routinely conflated

(i) **Statistical / semantic similarity.** The default embedding objective — contrastive or masked-language pretraining, then a cosine/dot-product similarity — produces a *symmetric, undirected* notion of relatedness: $\text{sim}(a, b) = \text{sim}(b, a)$. Two documentation steps that mention the same dataset, schema, or UI panel land near each other regardless of which must happen first. This is exactly the shortcut our project must defeat: in the repo’s hardest ordering dataset (`ajo_newstyle`), the best symmetric similarity model collapses to $\approx 65\%$ accuracy because topical overlap and step order are decorrelated on purpose. Semantic similarity is a *necessary* substrate (it gives a strong stage-1 retriever) but is provably insufficient for ordering: any symmetric kernel assigns the pair (A, B) and (B, A) identical scores and therefore carries zero bits about direction.

(ii) **Asymmetric / directional relations.** The next notion drops symmetry without yet claiming causality. Entailment is the canonical example: “ P entails H ” does not imply “ H entails P ” [19, 179]. Order embeddings, hyperbolic/entailment cones, and asymmetric dual encoders (a query tower and a document tower with untied weights) all realize a scoring function $s(a \rightarrow b) \neq s(b \rightarrow a)$. Temporal precedence (“ A occurs before B ”) and prerequisite structure also live here. Crucially, *asymmetry is not causality*: an asymmetric relation can encode narration order, set inclusion, or specificity, none of which is a causal mechanism. Most so-called “causal embeddings” in practice are this — a learned directional relation — and the honest claim stops here.

(iii) **Genuine causal structure.** The strongest notion requires that the represented relation track an underlying data-generating mechanism: $A \rightarrow B$ means intervening on A changes the distribution of B , while intervening on

B does not change A [131, 133]. This is defined with respect to a structural causal model (SCM) and is identified, in the observational regime, only under assumptions (Sec. 2.4). For text, we essentially never observe interventions on documentation steps, so genuine causal structure is at best *inferred* from the asymmetries the text exposes. Our position: the project targets notion (ii) as a tractable proxy for notion (iii), and the contribution is making that proxy as faithful as the text allows.

Problem statement. Formally, given an ordered pair of texts (t_1, t_2) we want a model $f_\theta(t_1, t_2) \rightarrow [0, 1]$ estimating $P(t_1 \text{ causally precedes / enables } t_2)$, with the constraints (a) *antisymmetry*, $f_\theta(t_1, t_2) \approx 1 - f_\theta(t_2, t_1)$ for genuinely directional pairs, and (b) an *abstain / NEUTRAL* option for independent or parallel steps so that the induced relation is a partial order (DAG), not a forced total order. Downstream we need this primitive to (1) classify direction, (2) rerank follow-ups, and (3) sort a workflow. A pure bi-encoder embedding cannot satisfy (a) by construction unless direction is injected via untied towers or an asymmetric scoring head; this motivates the cross-encoder / interaction-based designs surveyed later.

2.2 Causal representation learning: what it gives and does not give for text

The modern program of *causal representation learning* (CRL) [151] seeks to recover latent variables that correspond to causal factors of variation and to learn the causal graph among them, rather than entangled statistical features. Three pillars are relevant.

Independent causal mechanisms (ICM). The ICM principle holds that the causal generative process factorizes into autonomous modules $P(X) = \prod_i P(X_i \mid \text{pa}_i)$ whose mechanisms are independent: changing one mechanism (an intervention) leaves the others invariant [133, 151]. This *invariance* is the deepest reason causal features generalize across distribution shift, and it underlies invariant-risk and domain-generalization objectives [9, 67]. For our setting, ICM is aspirational: it suggests that a step’s *enabling preconditions* should be an invariant module reusable across workflows, which is precisely the generalization we need for far-apart (non-adjacent) step pairs.

Disentanglement and identifiability. A central negative result is that disentangled representations are *not identifiable* from i.i.d. observations without inductive biases or weak supervision [102]: infinitely many latent factorizations explain the same marginal. CRL therefore leans on auxiliary signals — multiple environments, interventions, temporal structure, or paired data — to pin down the causal factors [151]. **What this means for text:** we have no controlled interventions on documenta-

tion, but we *do* have weak supervision (the antisymmetric forward/reverse labels, multiple documents, tier metadata). The identifiability literature warns that direction learned from a single narration order is under-determined — it can latch onto narration rather than mechanism — which is exactly the failure diagnosed in the repo data (training only on adjacent, same-document pairs teaches narration order, not causal necessity). CRL thus does *not* hand us a causal text embedding for free; it tells us which auxiliary signals (non-adjacent pairs, multiple sources, abstention) are needed to make direction identifiable.

2.3 Pearl’s ladder and where a documentation “causal embedding” sits

Pearl’s *ladder of causation* stratifies causal questions into three rungs [131, 132]:

Rung 1 — Association $P(y \mid x)$: “seeing.” Correlation, conditional probability, similarity. All standard embeddings live here.

Rung 2 — Intervention $P(y \mid \text{do}(x))$: “doing.” The effect of actively setting x , requiring an SCM or experiments.

Rung 3 — Counterfactual $P(y_x \mid x', y')$: “imagining.” What would have happened had x been different, holding the realized world fixed.

An honest placement. A causal embedding for procedural documentation operates predominantly at *Rung 1 with structural / directional precedence layered on top*. We are estimating an associational quantity — $P(t_1 \text{ precedes } t_2 \mid t_1, t_2)$ — that we *interpret* as enablement. We are not running $\text{do}(\cdot)$ on a workflow, and we are not computing counterfactual re-orderings of a tutorial. The directional supervision (forward/reverse labels) nudges the representation toward the asymmetry that a Rung-2 mechanism would induce, but the trained object remains a Rung-1 estimator of a directed relation. This is the single most important caveat in the survey: claiming Rung-2/3 content for a text embedding trained on observational documentation would be an overclaim. Where genuine intervention-style evidence exists in our data, it is implicit (“you cannot publish before you authenticate” is a logical/temporal precondition recoverable from text, not an experiment).

2.4 Causal direction identification (background, then pivot to text)

Outside NLP, the *bivariate* causal-direction problem — given samples of (X, Y) , decide $X \rightarrow Y$ vs. $Y \rightarrow X$ — is the cleanest analog of our pairwise task. Because the joint $P(X, Y)$ is symmetric, direction is identifiable only under structural assumptions that break the symmetry:

- **Additive Noise Models (ANM)** [75]: assume $Y = f(X) + N$, $N \perp X$. A nonlinear f typically admits

an independent-noise fit in only one direction; one regresses both ways and tests residual independence (HSIC) to pick the direction with independent residuals.

- **LiNGAM** [156]: linear non-Gaussian acyclic models identify direction via independent-component analysis on the non-Gaussian noise.
- **IGCI** [79]: information-geometric causal inference assumes independence between the cause distribution and the mechanism, scoring an asymmetry in their “alignment.”

The standard testbed is the *Tübingen cause-effect pairs* benchmark of ≈ 100 expert-labeled real bivariate pairs [113]; reported accuracies cluster around $\sim 60\text{--}72\%$ for IGCI/ANM-style methods — i.e. direction is hard even with clean numeric data and explicit modeling assumptions. **The pivot to text:** documentation gives us neither i.i.d. numeric samples nor a clean noise model. Instead, the asymmetry-revealing signal is *linguistic* — discourse connectives, temporal markers, prerequisite phrasing, and world knowledge — which is why the rest of this survey treats causal direction in text as a supervised pairwise classification / entailment problem (Sec. 4) rather than an ANM/IGCI estimation problem. The bivariate literature contributes the conceptual lesson, not the algorithm: direction must come from an assumption or supervision that breaks the symmetry of $P(X, Y)$, and for text that assumption is encoded in the antisymmetric labels and the cross-attention architecture.

2.5 A taxonomy of “causal” embeddings

Table 1 organizes the design space along two axes: the *relation type* the model can express (symmetric, directional, or mechanism-grounded) and the *architecture* that realizes it. The taxonomy makes explicit that genuine causal grounding is not achievable by geometry alone in the observational text regime; the best a feasible model does is a faithfully directional, abstention-capable estimator (row 2–3), built on a strong symmetric retriever (row 1).

Relevance to causal embedding for AEP/AJO. This foundation fixes the target and its honest ceiling. The AEP/AJO project needs row 2–3 of Table 1: a *directional, abstention-capable* precedence estimator built on a strong symmetric retriever (row 1, where fine-tuned BGE-large already gives the best MRR in the repo). Three foundational lessons carry forward. First, symmetric similarity cannot order steps — direction must be *injected*, motivating cross-attention backbones over independent embeddings. Second, the identifiability results [102, 151] explain *why* training on adjacent same-document pairs fails to generalize: direction learned from a single narration order is under-determined, so we must supply the missing auxiliary signal (non-adjacent transitive pairs,

multiple sources, an abstain class). Third, we should advertise the model as a Rung-1 structural-precedence estimator, not an interventional one — the ICM/invariance ideal (row 4) is the north star for generalization, but genuine Rung-2/3 content is not recoverable from observational documentation. The next section turns to how direction is actually learned and evaluated from text.

3 Text Embeddings and Neural Retrieval: State of the Art

Neural text retrieval rests on learned dense representations of text that can be compared by an inexpensive geometric operation. This section surveys the architectural families that dominate the field, the contrastive objectives that train them, the recent wave of instruction-tuned and LLM-based embedders, and the representation-geometry choices (dimensionality, pooling, Matryoshka nesting) that govern their deployment. Throughout we tie the landscape back to the project’s own bi-encoder/cross-encoder findings on Adobe Experience Platform (AEP) and Adobe Journey Optimizer (AJO) workflow data, and flag the one assumption—*symmetric* cosine similarity—that makes all of these models a poor fit for directional, causal ordering.

3.1 Architecture families

Bi-encoders (dual-encoders). A bi-encoder maps a query q and a document d *independently* to vectors $\mathbf{e}_q, \mathbf{e}_d \in \mathbb{R}^k$ and scores them with a symmetric kernel, typically inner product or cosine: $s(q, d) = \mathbf{e}_q^\top \mathbf{e}_d$. Because document vectors can be precomputed and indexed, retrieval reduces to (approximate) nearest-neighbour search and scales to billions of passages. DPR [85] established the modern recipe of two BERT towers trained with in-batch negatives; Sentence-BERT [143] popularised siamese/triplet fine-tuning for sentence-level similarity; and SimCSE [62] showed that even unsupervised dropout-as-augmentation contrastive training yields strong sentence embeddings while reducing the anisotropy of raw BERT representations [96]. The repo’s **DirectionalClassifier** is a bi-encoder variant: two untied encoders produce CLS embeddings that are L2-normalised and fused as $[\mathbf{a}; \mathbf{b}; \mathbf{a}-\mathbf{b}; \mathbf{a} \odot \mathbf{b}]$ before an MLP head—the explicit $\mathbf{a}-\mathbf{b}$ term is an attempt to inject the asymmetry that plain cosine lacks.

Cross-encoders (rerankers). A cross-encoder concatenates both texts into a single input $[\text{CLS}] t_1 [\text{SEP}] t_2$ and lets full self-attention model token-level interactions, emitting a scalar relevance from the pooled state [125]; sequence-to-sequence variants such as monoT5 [127] cast ranking as generating a “true”/“false” token. Cross-encoders are far more accurate but cost a full forward pass *per pair*, so they are used only as second-stage rerankers

Table 1: Taxonomy of embedding/relation models for procedural documentation. “Pearl rung” is the highest rung the model honestly addresses in the observational text regime. Direction can be *injected* architecturally (untied towers, cross-attention) or via the training signal (antisymmetric labels).

Relation type	Typical objective	Symmetry	Architecture / realization	Pearl rung
Semantic similarity	Contrastive (InfoNCE), MLM	Symmetric	Single-tower / tied bi-encoder; cosine score	Rung 1 (assoc.)
Directional (learned)	Pairwise BCE on forward/reverse labels; entailment	Antisymmetric	Untied dual encoder; order/cone embeddings; cross-encoder over $[t_1; t_2]$	Rung 1 + directional precedence
Directional + abstain	As above + NEUTRAL class; partial-order/DAG decoding	Antisymmetric, partial	Cross-encoder + 3-way head; rank aggregation (Bradley–Terry, min-feedback-arc-set)	Rung 1 + structural precedence
Mechanism-grounded (aspirational)	Invariance / ICM; counterfactual consistency	Asymmetric, invariant	CRL latent SCM; reasoning trace with grounded premises (VeriCoT-style)	Rungs 2–3 (<i>not</i> attained from text alone)

over a shortlist from a bi-encoder.

Late interaction (ColBERT family). ColBERT [90] occupies a middle ground: it stores *per-token* embeddings and scores a pair by a sum of max-similarities (“MaxSim”), $s(q, d) = \sum_{i \in q} \max_{j \in d} \mathbf{e}_{q_i}^\top \mathbf{e}_{d_j}$, preserving fine-grained interaction while keeping the document side precomputable. ColBERTv2 [148] adds residual compression and denoised, distilled supervision, cutting the index footprint 6–10×, and PLAID [147] introduces centroid-based pruning that lowers late-interaction latency by up to 7× on GPU and 45× on CPU at equal quality.

The accuracy/latency tradeoff and the repo finding. The canonical wisdom—cross-encoders are more accurate, bi-encoders are faster—is borne out and *sharpened* by the project’s experiments. On pairwise *classification* (`aep_causal_cls34`), the bi-encoder *beat* the cross-encoder by roughly 4.6% (e.g. BGE-base 86.1% bi vs. 81.5% cross; MiniLM 87.4% vs. 78.8%), because the cross-encoder shares a single 512-token budget across both texts and is *starved* of context when the two steps are long. Yet on the task we actually care about—end-to-end *workflow ordering*—the DeBERTa-v3-large cross-encoder scored 21/30 while the bi-encoder scored 0/30. The reason is distributional: training pairs are same-document and windowed (adjacent steps, gap $\leq 2-4$), but ordering a full workflow requires judging *non-adjacent*, *off-distribution* pairs. The bi-encoder’s independent embeddings collapse there—two far-apart steps look semantically similar and cosine cannot tell which precedes—whereas the cross-encoder’s *joint cross-attention* reads the two texts *together* and can pick up directional cues (prerequisite mentions, references to prior configuration) that survive the distribution shift. Pairwise accuracy is therefore *not* a proxy for ordering quality.

3.2 Contrastive training and negatives

Dense encoders are trained almost universally with the InfoNCE / softmax-contrastive loss [165]. For a query q with one positive d^+ and negatives $\{d_j^-\}$,

$$\mathcal{L}_{\text{InfoNCE}} = -\log \frac{\exp(s(q, d^+)/\tau)}{\exp(s(q, d^+)/\tau) + \sum_j \exp(s(q, d_j^-)/\tau)},$$

where τ is a temperature controlling how sharply the model separates positives from negatives (small τ emphasises the hardest negatives). *In-batch negatives* reuse other examples in the minibatch as cheap negatives, so larger batches directly improve representation quality. The dominant lever, however, is *hard-negative mining*: negatives that are lexically or semantically close to the positive but actually irrelevant. ANCE [183] periodically re-mines hard negatives from the model’s own evolving ANN index; RocketQA [140] combines cross-batch negatives, denoised hard negatives (filtered by a cross-encoder to remove false negatives), and data augmentation; and TAS-B [73] balances topic-aware sampling with cross-encoder/ColBERT distillation. This directly motivates the repo’s `hard_neg_semantic` dataset ($\sim 261\text{K}$ pairs): forcing the model to distinguish a true causal successor from a topically identical but causally wrong step is exactly the hard-negative regime, and it is what pushes the encoder past the “same-topic” shortcut toward genuine directional discrimination.

3.3 Instruction-tuned and LLM-based embeddings

The current frontier replaces BERT-scale encoders with instruction-conditioned and decoder-LLM backbones. E5 [171] pre-trains on weakly supervised text pairs (CCPairs) and was the first to beat BM25 zero-shot on BEIR; its multilingual variant underpins much of the open ecosystem. INSTRUCTOR [158] prepends a natural-language *task instruction* so one model serves many tasks, and GTE [98] uses multi-stage contrastive training over di-

verse sources. The BGE family [182] and BGE-M3 [28] add self-knowledge distillation and, in M3, *multi-functionality* (dense, sparse, and multi-vector retrieval from one model), 100+ languages, and 8192-token context. The LLM-backbone line is now state of the art: E5-Mistral [173] fine-tunes Mistral-7B on GPT-generated synthetic retrieval tasks; SFR-Embedding-Mistral [110] extends it with multi-task transfer; LLM2Vec [15] converts any decoder-only LLM into an encoder via bidirectional attention plus masked-next-token and contrastive adaptation; NV-Embed [95] adds latent-attention pooling and two-stage instruction tuning to reach 72.31 on MTEB; GritLM [118] unifies generation and embedding in one model via instruction routing; and Qwen3-Embedding [198] (0.6B/4B/8B) tops the MMTEB multilingual leaderboard (70.58 for the 8B as of mid-2025), with even the 0.6B model competitive—directly relevant under our <1B budget.

What the leaderboards measure—and don’t. MTEB [117] aggregates ~56 tasks (retrieval/BEIR [164], clustering, classification, STS, reranking, pair-classification, summarisation) into a single average; MMTEB [51] extends this to 500+ tasks and 250+ languages with long-document, code, and instruction-following tasks. Crucially, *none* of these tasks is *directional* or *causal*: STS and pair-classification are symmetric ($\text{sim}(a, b) = \text{sim}(b, a)$), retrieval is topical relevance, and reranking judges relevance, not precedence. A model can top MMTEB and still be unable to tell whether step *A* enables step *B* or vice versa. This is borne out by the project’s own GritLM-7B probe: zero-shot, it *stays semantic*—it ranks by topical similarity and *ignores* a causal instruction in the prompt, never overriding its semantic bias in favour of directional precedence.

3.4 Representation geometry: dimensionality, pooling, Matryoshka

Pooling. How token states become one vector matters. CLS pooling reads a single special token (natural for BERT bi-encoders and the repo’s classifier); mean pooling averages token states (the default for E5/Sentence-BERT and usually more robust); last-token pooling reads the final position and is standard for causal decoder-LLM embedders (E5-Mistral, NV-Embed, Qwen3) because only that position has attended to the whole sequence—NV-Embed’s latent-attention pooling is an explicit fix for the weakness of plain last-token pooling.

Dimensionality and Matryoshka. Larger k (LLM embedders emit 1024–4096-d) helps quality but inflates index size and search cost. Matryoshka Representation Learning [93] trains nested prefixes so a single vector can be *truncated* (e.g. 1024 $\rightarrow$ 256 $\rightarrow$ 64) with graceful degradation, enabling an adaptive accuracy/latency knob and cheap coarse-to-fine retrieval; it is now standard in com-

mercial APIs and in BGE/Qwen3 releases. For a two-stage AEP/AJO system, MRL allows a small truncated vector for stage-1 recall and the full vector (or a reranker) for precision.

The symmetry mismatch. Every model above scores pairs with cosine or inner product, which is *symmetric*: $s(a, b) = s(b, a)$. A retrieval relevance signal is naturally symmetric, but causal precedence is *antisymmetric*—if *A* precedes *B* then *B* does not precede *A*. No amount of contrastive fine-tuning of a symmetric kernel can represent a strict order, which is precisely why the repo’s bi-encoder bolts on an **a–b** feature and why the cross-encoder (asymmetric by input order) wins on ordering. This is the central geometric gap addressed later (see the section on asymmetric geometry), which considers order embeddings, asymmetric/dot scoring, and hyperbolic/cone representations that natively encode direction.

Relevance to causal embedding for AEP/AJO

Three implications follow for the project. (1) A symmetric-cosine bi-encoder is an excellent *stage-1 recaller*—fine-tuned BGE-large already gives +21% MRR over the BERT baseline on AEP retrieval—but its independent, symmetric embeddings *cannot* order non-adjacent workflow steps, matching the 0/30 ordering result. (2) The cross-encoder’s joint attention is what generalises to off-distribution pairs (21/30); the right architecture is therefore a bi-encoder retriever feeding an *asymmetric, cross-attentive* reranker/orderer. (3) Instruction-tuned LLM embedders are tempting under the <1B budget (Qwen3-Embedding-0.6B is strong), but the GritLM finding warns that instructions alone do not buy directionality—these models optimise the symmetric, topical objective that MTEB/MMTEB reward, with no causal or directional task in sight. Closing that gap requires hard causal negatives (§3.2), an antisymmetric scoring geometry, and cross-attention, not merely a bigger embedder.

4 Causal Reasoning and Direction from Text

Section 2 established that direction in text must be *supervised* or *architecturally injected* rather than estimated from an assumed noise model. This section surveys the benchmarks and methods that do so: commonsense and formal causal-direction benchmarks, the natural-language inference (NLI) substrate that supplies directional precedence, event-causality and temporal-ordering corpora, the classifier families used to predict direction, and the consistent empirical finding that *causal direction is hard* even for frontier models. We close by connecting these to the AEP/AJO ordering task.

4.1 Commonsense and formal causal benchmarks

Commonsense causal choice. COPA (Choice of Plausible Alternatives) [145] is the canonical directional benchmark: 1,000 forced-choice items (500 dev / 500 test), each giving a premise and two alternatives for *either the cause or the effect*, with the prompt itself specifying direction (“What was the CAUSE?” vs. “What happened as a RESULT?”). COPA is part of SuperGLUE [169]; the human baseline is 100 and a 1.5B-parameter DeBERTa reaches 98.4 on COPA, so the *benchmark* is near-saturated while the underlying *direction skill* (below) is not. **e-CARE** [47] scales this to “over 21K” causal-reasoning questions, each paired with a free-text conceptual explanation. The best model (ALBERT) reaches 73.86% vs. human 92.00%, and — the key result for our project — pretraining on e-CARE explanations *transfers*: COPA improves 70.4 $\rightarrow$ 75.4 and EventStoryLine causality F1 66.5 $\rightarrow$ 68.1. Explaining causality, not merely labeling it, measurably helps downstream causal tasks.

If-then commonsense knowledge. ATOMIC [149] is a crowd-sourced knowledge graph of ≈ 877 K if-event-then triples over nine typed relations that explicitly distinguish causes from effects, agents from themes, and voluntary from involuntary events — a directed inferential resource directly analogous to “effects of step A” / “preconditions of step B.”

Formal causal inference. Corr2Cause [82] is the first benchmark for pure causal inference (inferring causation from correlation), with >200 K samples over 17 LLMs. The headline is stark: LLMs are near-random, with GPT-4 F1 = 29.08 (precision 20.92 / recall 47.66 / accuracy 64.60; the high accuracy is a class-imbalance artifact, so F1 is the honest metric). On direction specifically, under variable refactorization the Is-Ancestor / Is-Descendant F1 drop to 45.45 / 29.41, and non-finetuned GPT-4 “fails at all direct causation recognition with parents and children” — it cannot reliably tell $A \rightarrow B$ from $B \rightarrow A$. CLADDER [81] grounds 10K questions in Pearl’s three rungs with oracle ground truth from a causal-inference engine; its *CausalCoT* prompting lifts GPT-4 from 62.03% $\rightarrow$ 70.40% (random 49.27%), a +8.37-point gain, with by-rung accuracy of associational 83.35%, interventional 67.47%, counterfactual 62.05% — monotonically harder up the ladder. CRASS [57] tests counterfactual reasoning via “questionized” counterfactual conditionals against a crowd-validated human baseline (part of BIG-bench); the tested LLMs trail humans by over 25 points. CaLM [30] is the broadest recent effort: 126,334 samples, 92 causal targets covering 46 tasks across natural, symbolic, and mathematical modes and two languages, evaluating 28 models — a taxonomy (causal target / adaptation / metric / error) that confirms the same pattern, with direction- and intervention-style targets the

weakest.

4.2 NLI / entailment as a substrate for directional precedence

NLI is inherently *directional* — “premise entails hypothesis” is not symmetric — which makes it a natural template for “step A precedes / enables step B.” The foundational corpora are SNLI (≈ 570 K human-written pairs) [19] and MultiNLI (≈ 433 K, ten genres, defining the matched/mismatched split) [179]. ANLI [123] hardens this with an adversarial human-and-model-in-the-loop procedure (R1/R2/R3 train sizes 16,946 / 45,460 / 100,459), so a credible NLI score must be reported on adversarial data. DeBERTa-v3 cross-encoders are the strong, cheap NLI workhorse: a DeBERTa-v3-large checkpoint trained on ≈ 885 K NLI pairs reaches 0.912 / 0.908 matched/mismatched MNLI accuracy and 0.702 average on ANLI [71, 94]. The cross-encoder lets every premise token attend to every hypothesis token — precisely the interaction a directional judgment needs, and the architectural reason NLI leaderboards are topped by cross-encoders rather than bi-encoders.

Document-level and explanation-augmented NLI.

DocNLI [187] reformats QA and summarization into a large document-granularity NLI dataset (premise always document-length, hypotheses from a sentence to a passage), relevant because documentation steps are multi-clause passages, not single sentences. e-SNLI [21] adds ≈ 549 K human explanations and rationale highlights, enabling models that justify the entailment, while EntailmentBank [39] provides 1,840 multistep entailment trees — the template for proving “A enables B” through intermediate steps. INLI [70] draws the distinction most relevant to us: *explicit* entailment (lexical/syntactic) vs. *implied* entailment (requiring world knowledge / logical reasoning). On 10K premises with 40K hypotheses, a model trained on INLI reaches 92.4% overall and 88.5% on implied cases, while models trained on SNLI/MNLI/ANLI/WANLI score only $\approx 50\%$ (near chance) on implied entailments. Causal precedence (“you must authenticate before you can publish”) is almost always *implied*, so surface NLI models are near-chance on exactly the regime we care about — the central argument for injecting reasoning rather than relying on similarity.

Verifying the reasoning chain. VeriCoT [56] formalizes each chain-of-thought step into first-order logic, identifies whether each premise is grounded in source context / commonsense / a prior step, and uses automated solvers to check logical validity; evaluated on ProofWriter, LegalBench, and BioASQ it is “a strong predictor of final-answer correctness.” This is a template for validating a model’s “A precedes B because...” trace — flagging ungrounded (hallucinated) precedence claims against the documentation.

4.3 Event causality and temporal ordering from text

The corpora most directly analogous to ordering workflow steps annotate relations *between events* in text. **TimeBank** [139] introduced TimeML event/temporal-link annotation; **TimeBank-Dense** [25] added dense (near-complete) pairwise temporal ordering. **MATRES** [124] simplified TimeBank-Dense via a multi-axis scheme annotating only start-points into four relations (BEFORE, AFTER, EQUAL, VAGUE); reported systems reach ≈ 69 – 75.5 F1 on MATRES vs. ≈ 48 – 64.5 on TB-Dense, so even temporal ordering — strictly easier than causal ordering — is far from solved. On the causal side, **Causal-TimeBank** [112] annotates causality over the TempEval-3 corpus (184 docs, 6,813 events, only 318 of 7,608 event pairs causal — extreme sparsity), and **EventStoryLine** [24] provides a larger benchmark for joint causal-and-temporal relation extraction (22 topics, 258 docs, $\approx 1,770$ intra- and 3,885 inter-sentence causal pairs). **CaTeRS** [115] jointly annotates temporal (BEFORE/AFTER) and causal (CAUSE/ENABLE/PREVENT) relations over 1,600 sentences, explicitly separating “*A* before *B*” from “*A* enables *B*.” **This temporal-vs-causal distinction is exactly the trap in our data:** narration order in a tutorial is not the same as causal necessity, and a model trained on positional order alone learns the former.

4.4 Methods for causal / temporal direction classification

Three operationalizations of direction recur [74]. (1) *Pair-wise span/relation classifiers*: given two spans known to be in a causal relation, predict cause $\rightarrow$ effect vs. effect $\rightarrow$ cause. On PDTB3, BERT/SpanBERT reach 0.83/0.86 F1; on EventStoryLine, 0.79/0.71 — i.e. supervised direction prediction caps at ~ 0.79 – 0.86 F1 even with the relation given. The same work builds **CREST**, unifying nine causal datasets. (2) *Directed-graph edges*: **CausalBank** [99] (≈ 314 M cause-effect pairs split into cause-pattern-effect / effect-pattern-cause, encoding direction lexically) and **CauseNet** [72] (a directed web-extracted causality graph) provide direction as edge orientation. (3) *Joint / direction-aware models*: a recent ECI survey [205] notes prior work “mainly focus[es] on causality existence but ignore[s] causal direction,” and *Identifying while Learning* [29] jointly learns existence and direction, exploiting that causal chains are *asymmetric and transitive* — the same transitivity our rank-aggregation step relies on.

4.5 Why direction is hard; frontier vs. small models

The evidence converges on a single conclusion: causal *direction* is the consistently hard part of causal NLP, and scale does not fix it. GPT-4 is near-random on Corr2Cause (F1 29.08) and cannot reliably distinguish

ancestor from descendant [82]. *Is ChatGPT a Good Causal Reasoner?* [60] shows ChatGPT/GPT-4 are *outperformed by fine-tuned small models* on event-causality identification, with high recall but low precision — a *reporting-bias* artifact (text states what causes what, rarely what does *not*), so the model over-predicts causality and is worse on *implicit* (reasoning-required) than explicit (cue-word) causality, and degrades as event density rises. Supervised direction classifiers cap at 0.79–0.86 F1 [74]; even clean numeric bivariate methods top out at ~ 60 – 72% on the Tübingen pairs [113]. The reasons are structural: (a) the joint distribution is symmetric so direction needs an extra assumption or supervision (Sec. 2); (b) the relation is usually *implied*, where surface NLI is near-chance [70]; (c) labels are sparse and imbalanced [112]. What *does* help is structured reasoning: Causal-CoT’s +8.37 on CLADDER [81] and e-CARE explanation transfer [47] both show that a causal-specific reasoning step, not a larger embedding, moves the needle.

Relevance to causal embedding for AEP/AJO. Mapped to our pairwise primitive $P(t_1 \text{ precedes } t_2)$, the literature gives a concrete recipe and concrete warnings. First, frame the task as *directional NLI*: an ordered pair (t_1, t_2) with labels PRECEDES / FOLLOWS / NEUTRAL, evaluating both directions of the same pair to expose the symmetry failure that Corr2Cause flags — and use a *cross-encoder* (DeBERTa-v3-style) so tokens of both steps interact, matching the repo finding that the cross-encoder orders far better (21/30) than the bi-encoder (0/30) despite lower pairwise accuracy. Second, keep the NEUTRAL/abstain class so parallel steps are not forced into a chain, turning the output into a DAG amenable to robust rank aggregation (consistent with the transitivity exploited by direction-aware ECI). Third, separate *temporal* from *causal* precedence (the CaTeRS lesson): our positional labels teach narration order, so non-adjacent / cross-document pairs and explanation-augmented training (the e-CARE transfer effect) are needed to recover genuine enablement. Fourth, because direction is hard and *implied*, inject reasoning — distilled CausalCoT-style traces filtered by the verifiable gold label, with optional VeriCoT-style grounding checks — rather than relying on similarity. The honest ceiling remains a Rung-1 structural-precedence estimator (Sec. 2.3), but the benchmarks show that a directional, abstention-capable, reasoning-augmented cross-encoder is the design most likely to close the 0/30 ordering gap.

5 Asymmetric and Order-Structured Embeddings for Directionality

Causal precedence is an inherently *asymmetric, transitive* relation: if step A enables step B then B does not enable A , and if $A \prec B$ and $B \prec C$ then $A \prec C$. The dominant tool for dense retrieval and sentence similarity — cosine similarity — cannot express either property. This section surveys the geometric machinery that can: order embeddings, density and box embeddings, hyperbolic geometry, and asymmetric scoring heads. We connect each family to the project’s own experiments, in particular the **internship-causal-embedding-lorentz** subproject, whose Lorentzian space/time encoder is a direct instantiation of several of these ideas.

5.1 Why cosine similarity is the wrong tool

For unit-normalized embeddings $\mathbf{u}, \mathbf{v}$, cosine similarity $s_{\cos}(\mathbf{u}, \mathbf{v}) = \mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ is *symmetric* by construction: $s_{\cos}(\mathbf{u}, \mathbf{v}) = s_{\cos}(\mathbf{v}, \mathbf{u})$. Any scorer that factors through a single shared metric space and reads out an inner product or a distance $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ is likewise symmetric and obeys the triangle inequality, so it can encode “ A and B are about the same topic” but never “ A causally precedes B .” Empirically this is exactly what the repo observed: a pure space-only cosine scorer has directional accuracy $\equiv 0.5$ by construction [26]. To model precedence we must abandon symmetry and adopt a score $s(A, B) \neq s(B, A)$ whose asymmetry is a controllable, learnable quantity. We organize the asymmetric mechanisms into four geometric families plus a class of learned asymmetric operators.

5.2 Order embeddings (partial order via reversed product order)

Vendrov et al. [166] embed objects in the non-negative orthant $\mathbb{R}_+^N$ and impose the *reversed product order*: $\mathbf{x} \preceq \mathbf{y}$ (read “ $\mathbf{y}$ entails / is more general than $\mathbf{x}$ ”) iff $x_i \geq y_i$ for every coordinate i . The apex of the hierarchy (most general concept) sits near the origin; specializations move outward. Order violation is measured by the penalty

$$E(\mathbf{x}, \mathbf{y}) = \|\max(\mathbf{0}, \mathbf{y} - \mathbf{x})\|_2^2, \quad (1)$$

which is 0 exactly when $\mathbf{x} \preceq \mathbf{y}$ holds and grows with the violation otherwise — and is *asymmetric*, $E(\mathbf{x}, \mathbf{y}) \neq E(\mathbf{y}, \mathbf{x})$. Training uses a max-margin objective over positive ordered pairs P and negative (non-ordered) pairs N :

$$\mathcal{L} = \sum_{(\mathbf{x}, \mathbf{y}) \in P} E(\mathbf{x}, \mathbf{y}) + \sum_{(\mathbf{x}, \mathbf{y}) \in N} \max(0, \alpha - E(\mathbf{x}, \mathbf{y})). \quad (2)$$

Crucially the construction is *transitive* by lattice structure, matching the transitivity of causal precedence, and

it scales: scoring is one coordinate-wise comparison, so order embeddings remain indexable like ordinary dense retrieval.

5.3 Probabilistic and density order embeddings

A complementary route makes each object a *distribution* rather than a point. Vilnis and McCallum [167] represent each word as a Gaussian $\mathcal{N}(\boldsymbol{\mu}_w, \Sigma_w)$ and score relatedness by KL divergence,

$$\text{KL}(\mathcal{N}_a \parallel \mathcal{N}_b) \neq \text{KL}(\mathcal{N}_b \parallel \mathcal{N}_a), \quad (3)$$

whose built-in asymmetry encodes inclusion of one ellipsoid in another and captures entailment direction and uncertainty (broad covariance = general concept). Athiwaratkun and Wilson [10] extend this to *hierarchical density order embeddings*: they impose a partial order on Gaussians via a thresholded (“truncated”) divergence so that a general concept’s density *encapsulates* its specializations, and train with an order-violation loss analogous to Eq. (2). This unifies the order-embedding inductive bias with calibrated, interpretable uncertainty and gives strong WordNet hypernymy and HyperLex results.

5.4 Box / region-based embeddings

Box embeddings represent each concept as an axis-aligned hyper-rectangle (“box”) $\text{Box}(x) = \prod_i [x_i^-, x_i^+]$ and read off relations from *volume and intersection*, giving a probabilistic, set-theoretic semantics. Vilnis et al. [168] define a calibrated containment probability

$$P(b \mid a) = \frac{\text{Vol}(\text{Box}(a) \cap \text{Box}(b))}{\text{Vol}(\text{Box}(a))}, \quad (4)$$

which is asymmetric ($P(b \mid a) \neq P(a \mid b)$) whenever the boxes differ in volume) and directly models entailment/containment plus negative correlation between concepts. Boxes are closed under intersection, so conjunctive queries compose naturally — the basis of **query2box** [144], which embeds a knowledge-graph query as a box, realizes conjunction as box intersection and projection as a learned box transform, and ranks an entity by whether its point falls inside the answer box. A practical caveat is that hard-edged volumes give vanishing gradients when boxes are disjoint; smoothed/Gumbel box measures restore trainability [42].

5.5 Hyperbolic embeddings

Hierarchies and DAGs grow their node count *exponentially* with depth; Euclidean space, whose volume grows polynomially, cannot embed a tree with low distortion, but hyperbolic space (constant negative curvature) has volume that grows exponentially with radius and so embeds trees almost isometrically in few dimensions. Nickel and Kiela [121] place concepts in the Poincaré ball $\mathbb{B}^n =$

$\{\mathbf{x} : \|\mathbf{x}\| < 1\}$ with distance

$$d_{\mathbb{B}}(\mathbf{u}, \mathbf{v}) = \operatorname{arccosh}\left(1 + 2 \frac{\|\mathbf{u} - \mathbf{v}\|^2}{(1 - \|\mathbf{u}\|^2)(1 - \|\mathbf{v}\|^2)}\right), \quad (5)$$

so distances explode toward the boundary; depth (generality) is encoded by radius and similarity by angle. Nickel and Kiela [122] re-derive the same geometry in the Lorentz / hyperboloid model $\{\mathbf{x} : \langle \mathbf{x}, \mathbf{x} \rangle_{\mathcal{L}} = -1\}$ with the Minkowski form $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{L}} = -x_0 y_0 + \sum_{i \geq 1} x_i y_i$, which is numerically more stable for Riemannian optimization. Distance itself is still symmetric, so directionality is added on top: Ganea et al. [58] attach to each point a *geodesically convex entailment cone* and declare $\mathbf{x} \preceq \mathbf{y}$ iff $\mathbf{y}$ lies in the cone at $\mathbf{x}$, penalizing the angle by which a positive pair falls outside the cone — an asymmetric, transitive order over hyperbolic points. Ganea et al. [59] lift standard layers (linear maps, RNN/GRU, softmax) to the Poincaré ball via the exponential and logarithmic maps, enabling end-to-end hyperbolic encoders.

Connection to the repo’s Lorentz experiment. The `internship-causal-embedding-lorentz` subproject built exactly such an encoder: an e5/BGE backbone whose output is split into a *space* part $\mathbf{s} \in \mathbb{R}^{d_s}$ (L2-normalized, topical) and a *time* part $\mathbf{t} \in \mathbb{R}^{d_t}$ (Layer-Norm, unbounded, “causal clock”), scored by

$$s(a, b) = \alpha \cos(\mathbf{s}_a, \mathbf{s}_b) + \beta \frac{1}{d_t} \sum_{k=1}^{d_t} \sigma(\tau(t_{b,k} - t_{a,k})), \quad (6)$$

where the first term is symmetric (retrieval) and the second is an asymmetric SteepSigmoid time-ordering term ($\tau=10$) that is high only when b is “after” a in every time coordinate [26]. Two findings are decisive for this survey. (i) The time head genuinely breaks symmetry: with an explicit ordering loss it reaches **directional accuracy** 1.0 (forward > reverse for every test pair) versus 0.5 for the cosine baseline, on `aep_causal` and `followupqg`. (ii) The same time head contributes essentially *nothing* to retrieval MRR (e.g. $0.460 \rightarrow 0.459$ on `aep_causal` at $\beta=0$ vs 0.3, on identical pools): retrieval is dominated by topical similarity, and direction only says *which way* a known pair points, not *which* candidate is correct. The honest ablation further attributes the headline “+20% over the BERT baseline” to the *backbone swap* (BERT $\rightarrow$ e5), not to the Lorentzian machinery. The lesson: explicit directional geometry is the right tool for the *direction* sub-task but does not substitute for a strong retriever, and direction in this data lives in the *untied two-tower heads*, not in a single shared (even hyperbolic) clock coordinate.

5.6 Learned asymmetric operators and dual-space heads

Beyond geometric priors, one can simply *learn* an asymmetric scorer. **Dot product** (without normalization) is

already asymmetric in norm and underlies dense retrievers [86]. A **bilinear / relation matrix** $s(a, b) = \mathbf{u}_a^\top \mathbf{R} \mathbf{v}_b$ with non-symmetric $\mathbf{R}$ models a directional relation directly. Translational KG embeddings impose direction by construction: **TransE** [17] scores a triple by $-\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|$, so (h, r, t) and (t, r, h) generally differ; **RotatE** [162] makes $\mathbf{r}$ a unit-modulus rotation in $\mathbb{C}^d$, $\mathbf{t} = \mathbf{h} \circ \mathbf{r}$, which can represent *antisymmetry* (rotation by $\theta \neq 0, \pi$), inversion, and composition — precisely the algebra a precedence relation needs. **Dual-/two-space embeddings** give each text two roles via separate “cause” and “effect” projection heads, $\phi_c(\cdot)$, $\phi_e(\cdot)$, and score $s(A \rightarrow B) = \langle \phi_c(A), \phi_e(B) \rangle$; because $\phi_c \neq \phi_e$, $s(A \rightarrow B) \neq s(B \rightarrow A)$, and the two indexes remain separately retrievable. This is the same mechanism the repo found to carry direction: untied two-tower e5 with a directional margin loss pushing $s(a \rightarrow b) > s(b \rightarrow a) + m$ lifted directional accuracy to 0.91–1.00 on four of five datasets while keeping retrieval MRR $\geq 10\%$ above the BERT baseline [26].

The heads / fusion trick. The repo’s `DirectionalClassifier` encodes A and B separately and fuses them as $[\mathbf{a}; \mathbf{b}; \mathbf{a} - \mathbf{b}; \mathbf{a} \odot \mathbf{b}] \rightarrow \text{MLP}$. Read geometrically, this is a learned asymmetric operator: the difference term $\mathbf{a} - \mathbf{b}$ is *antisymmetric* ($(\mathbf{a} - \mathbf{b}) = -(\mathbf{b} - \mathbf{a})$), so it injects a TransE-like translational signal that flips sign under argument swap and lets the MLP read off direction; the Hadamard term $\mathbf{a} \odot \mathbf{b}$ is symmetric and supplies a bilinear-style interaction (topical agreement). Concatenating both gives the head a symmetric channel for *relatedness* and an antisymmetric channel for *precedence* — the fusion analogue of the space/time split in Eq. (7).

Relevance to causal embedding for AEP/AJO. An order/box/hyperbolic or dual-head scorer is attractive precisely where a cross-encoder cannot go: *scalable* stage-1 directional retrieval. All of the asymmetric families in Table 2 are single-vector indexable, so they can rank thousands of candidate next-steps in sublinear time while still respecting direction — something a symmetric cosine index cannot do and a cross-encoder cannot afford at corpus scale. The repo evidence says where this pays off and where it does not. *It pays off* when the direction is genuinely learnable from text and the workflow is close to a partial order/DAG: `qrecc`, `multiwoz`, and `followupqg` reached 0.91–1.00 directional accuracy with a dual-head + directional margin — a region-/order-style head would be a natural, transitivity-aware generalization there, and would let stage-1 itself prune wrong-direction candidates. *It will not help* on two fronts established by the project: (i) on `aep_causal`, where anchor and positive are near-duplicate documentation chunks, direction is ambiguous *in the text* and no geometry (towers, time head, cones) moves accuracy off ≈ 0.76 ; (ii) on the actual *ordering* metric, where pairwise classification accuracy does not

Table 2: Geometries for directional/causal scoring. “Indexable” = candidates can be embedded offline and retrieved with a single-vector ANN index (scalable stage-1); cross-encoders are not.

Family	Space	Asymmetry mechanism	Suited structure	Indexable / cost
Cosine bi-encoder	$\mathbb{S}^{d-1}$	none (symmetric)	topical similarity	yes / cheap
Order embeddings [166]	$\mathbb{R}_+^d$	reversed product order, $\max(0, \mathbf{y} - \mathbf{x})$	partial order, DAG	yes / cheap
Gaussian / density [10, 167]	Gaussians	asymmetric KL / truncated divergence	hypernymy + uncertainty	yes / medium
Box / region [144, 168]	boxes in $\mathbb{R}^d$	conditional volume $P(b a)$	containment, conjunctive queries	yes / medium
Hyperbolic + cones [58, 121]	$\mathbb{B}^d$ / hyperboloid	entailment cone membership	deep trees / DAGs	yes / medium
Translational KG [17, 162]	$\mathbb{R}^d$ / $\mathbb{C}^d$	$\mathbf{h} + \mathbf{r} - \mathbf{t}$, rotation	typed directional relations	yes / cheap
Dual-head (cause/effect)	two $\mathbb{R}^d$	$\phi_c \neq \phi_e$, learned	pairwise precedence	yes / cheap
Cross-encoder	joint	full cross-attention	adjacency / fine pairs	no / expensive

equal ordering quality and the DeBERTa *cross-encoder* (joint cross-attention, not indexable) is what generalizes to far-apart/non-adjacent pairs. The synthesis matching the rest of this survey is therefore a two-stage design: an asymmetric, indexable geometric head (dual cause/effect towers, optionally with an order/box penalty for transitivity) as a fast directional *retriever*, followed by a cross-encoder reranker for the hard adjacent pairs — reserving the expensive joint model for the few candidates where direction is genuinely hard.

6 Reasoning Rerankers and Reinforcement Learning for Ranking

A second-stage *reranker* re-scores a small candidate set returned by a first-stage retriever. With large language models (LLMs) this re-scoring can be framed as a generation problem, and a 2024–2025 wave of work has equipped these rerankers with explicit *reasoning* and trained them with reinforcement learning (RL). This section surveys the prompting paradigms, the reasoning rerankers, the RL recipes that make reasoning feasible at small scale, and the distillation techniques that internalise reasoning into a sub-1B (or even non-generative) scorer. Throughout, the target application is a *causal-precedence* scorer of < 1B parameters that judges whether one procedural step enables another, reranks candidate follow-ups, and feeds a workflow orderer (Sec. 9).

6.1 LLM reranking paradigms

The standard taxonomy distinguishes four prompting paradigms by *how many candidates enter a single prompt* and what the model emits, trading cost against effectiveness [200].

Pointwise. One query–document pair per call; the model emits an independent relevance score (via a yes/no token probability or a graded label). Cost is $O(N)$ for N candidates and the scores are *directly comparable and reusable*, but they require calibration (Sec. 9.4) and the per-item signal is the weakest of the four.

Pairwise. Two documents per call (“which is more relevant?”), aggregated into an order by sorting. This yields the strongest relative judgments but costs up to $O(N^2)$ comparisons, reduced to $O(N \log N)$ with sorting networks. Pairwise prompting maps naturally onto *causal precedence* (“does step A enable step B ?”), which is the primitive our project needs.

Listwise. A whole candidate list enters one prompt and the model emits a permutation using global context. RankGPT [161] established the sliding-window protocol (window 20, step 10 over a top-100 list, candidates labelled [1] [2] ... and reordered as [2] > [3] > [1]), and distilled GPT into open students — notably a 435M DeBERTa-v3-large student trained with a RankNet pairwise loss that reportedly surpassed the prior monoT5-3B state of the art on BEIR. RankVicuna [137] and RankZephyr [138] are open 7B listwise rerankers distilled from GPT-3.5/4 permutations; RankZephyr’s two-stage distillation (~100K RankGPT-3.5 then ~5K RankGPT-4 permutations) matches or exceeds RankGPT-4 and is robust to input order. Listwise is cheap per list (few calls) but suffers strong *position bias* and finite context.

Setwise. A set of $c+1$ documents enters each call and the model selects only the most relevant; plugged into heapsort/bubblesort this cuts sort complexity to $O(k \log_c N)$ [203]. On Flan-T5-large/DL19, setwise heapsort used roughly half the inferences of pairwise heapsort at comparable effectiveness. The “pick the most relevant of a set” objective is the unit on which Rank-R1 later builds its RL.

Injecting side signals. InsertRank [33] appends each candidate’s BM25 lexical score into a listwise prompt to ground reasoning on reasoning-intensive queries (BRIGHT, R2MED), giving consistent NDCG@10 gains with no fine-tuning. The lesson is that a cheap structural prior injected into the prompt measurably helps — analogous to injecting a bi-encoder similarity or a workflow-position prior into a causal reranker.

6.2 Reasoning rerankers

Reasoning rerankers apply the DeepSeek-R1 recipe [43] — `<think>/<answer>` formatting plus a verifiable relevance reward optimised by GRPO — to ranking.

Rank-R1. Zhuang et al. [204] take the Setwise prompt, add a reasoning instruction, and train the reasoning ability *purely with RL* on relevance labels (no reasoning supervision). On Qwen2.5 at 3B/7B/14B, GRPO uses a binary reward (1 iff the output matches the `<think>/<answer>` format *and* the selected label is the relevant document) and matches the Setwise SFT baseline using only $\sim 18\%$ of its data, with the largest gains on reasoning-intensive (BRIGHT) queries.

REARANK. Zhang et al. [195] train a *listwise* reasoning agent (REARANK-7B, on Qwen2.5-7B) with GRPO from only ~ 179 annotated MS MARCO queries (expanded by augmentation to $\sim 12\text{K}$ instances). Its composite reward $r = 0.8 r_{\text{rank}} + 0.1 r_{\text{fmt1}} + 0.1 r_{\text{fmt2}}$ uses a min-max-normalised NDCG@10 *improvement* ($\text{NDCG}_{\text{rerank}} - \text{NDCG}_{\text{init}} / (\text{NDCG}_{\text{opt}} - \text{NDCG}_{\text{init}})$) plus two format terms. It matches GPT-4 on TREC DL19/20 and surpasses it on BRIGHT, demonstrating that a listwise reasoning reranker — whose permutation output is exactly a step ordering — can be learned from a tiny label set.

Rank-GRPO. Zhu et al. [202] generalise GRPO for ranked outputs by treating each *rank position* (not each token or sequence) as the unit of advantage estimation and importance weighting, removing non-causal credit assignment and the quality decay toward the end of a generated list. Although framed for conversational recommendation, the rank-level credit assignment is directly relevant to any listwise reasoning reranker.

When chain-of-thought *hurts*: the think-free fix.

A central, initially counter-intuitive finding is that naive chain-of-thought (CoT) can *degrade* pointwise rerankers [7]. The mechanism: ranking needs a precise, calibrated *scalar* comparable across candidates, whereas free-form CoT (i) injects variance into the verbalised score, (ii) lets the model “overthink” and drift from the query, and (iii) consumes the output distribution that a clean score token would otherwise occupy — all at large latency cost. TFRank [54] resolves this with a *think-mode switch*: the model is *trained* with explicit CoT (internalising reasoning into its weights) but runs *think-free* at inference, emitting only a pointwise score with no reasoning chain. Built on the Qwen3 series with released checkpoints at **0.6B**, 1.7B, 4B and 8B, a *1.7B* TFRank competes with 7B reasoning rerankers on BRIGHT at $\sim 4\times$ fewer parameters and an order-of-magnitude faster inference. Its score is an expected-value over a five-level ordinal scale (0–4) read from the top-20 logit tokens, with a neutral 0.5 fallback

— precisely the role a cross-encoder classifier plays, but reasoning-distilled. The TFRank **0.6B** checkpoint is the existence proof that a sub-1B think-free reasoning scorer is achievable rather than speculative. Rank1 [178] occupies the opposite extreme (emit CoT at inference as test-time compute), underscoring the reason-then-rank versus reason-then-score latency trade-off.

6.3 Reinforcement learning for ranking and reasoning at sub-1B scale

GRPO and its core mechanism. Group Relative Policy Optimization [155], scaled in DeepSeek-R1 [43], removes PPO’s value network. For a prompt q it samples a group of G completions, scores each with a reward r_i , and assigns every token of completion i the group-relative advantage

$$A_i = \frac{r_i - \text{mean}(r_1, \dots, r_G)}{\text{std}(r_1, \dots, r_G)}, \quad (7)$$

optimised with PPO’s clipped surrogate plus a KL penalty to a frozen reference. The group mean replaces the learned critic, halving memory — attractive on constrained GPUs — but rollout cost scales with G , and groups that are all-correct or all-wrong yield zero advantage.

GRPO variants. *Dr. GRPO* [101] identifies two optimisation biases in Eq. 7: response-length normalisation (which inflates length, especially of wrong answers) and per-question std normalisation (which over-weights easy/hard prompts). It normalises by a fixed constant and drops the std term, cutting incorrect-response length by $\sim 38\%$ while preserving accuracy. *DAPO* [188] adds Clip-Higher (decoupled clip range $\varepsilon_{\text{low}}=0.2$, $\varepsilon_{\text{high}}=0.28$, fixing entropy collapse), dynamic sampling (discarding zero-gradient groups), token-level loss, and overlong-reward shaping; it removes the KL term and reaches AIME-2024= 50 on Qwen2.5-32B in half the training steps. *S-GRPO* [37] samples serial early-exit positions on a single chain with decaying reward, cutting sequence length 35–61% with no accuracy loss — useful when a binary causal decision should not invite runaway CoT.

PPO versus DPO versus GRPO. PPO [152] is actor-critic: a learned value baseline is lower-variance but unstable and doubles memory; it can edge out DPO on reasoning in controlled studies [78]. DPO [141] skips the RL loop and reward model, optimising a classification loss directly on (chosen, rejected) pairs — cheap and stable, but it does not explore. GRPO is the middle ground for *verifiable* reasoning (no reward model, no critic). Because the causal direction/order ground truth is verifiable, GRPO/Dr. GRPO is the natural fit; DPO over (correct order $\succ$ corrupted order) pairs is a cheaper baseline.

Reward design. The de-facto pattern is a sum of independent terms [43]: a *format* reward (+1 if the output matches the required schema) plus a *correctness* reward checked by a deterministic verifier. For binary causal direction the verifier is trivial (predicted=gold). For *ordering*, sparse exact-match wastes gradient, so order-sensitive rewards are preferred: REARANK’s NDCG-improvement [195], or a Kendall- τ /Spearman correlation between the predicted and gold permutation (dense in $[-1, 1]$), optionally plus an exact-permutation bonus and a format term. Dense, order-aware rewards matter at small scale because they avoid the all-wrong degenerate groups that give zero advantage.

Distill-then-RL, not RL-from-scratch. The strongest evidence for the sub-1B regime is that *distillation (SFT on reasoning traces) should precede any RL*. DeepSeek-R1 [43] reports that distilling a large model into a small dense model beats large-scale RL run directly on that small model, which “requires enormous computational power and may not even achieve the performance of distillation.” Open-RS [40] confirms the nuance: GRPO on 7K math samples lifted an *already-distilled* 1.5B model cheaply ($\sim \$42$, $4\times A40$), but observed degradation after 150–200 steps, KL instability and length truncation — small-model RL is fragile. Tina [174] reproduced a competitive 1.5B reasoning model with LoRA-GRPO for $\sim \$9$. “SFT Memorizes, RL Generalizes” [34] gives the canonical justification: RL with outcome reward generalises better out of distribution, *but SFT is still required first* to stabilise output format so RL pays off. A weak $< 1B$ base lacks the latent reasoning for RL to discover much unaided; seed it with SFT, then run a short, well-regularised GRPO polish (tens-to-low-hundreds of steps).

6.4 Distilling reasoning into small / non-generative models

Rationale-as-auxiliary-objective. Distilling Step-by-Step [76] trains a student with *two* objectives — predict the label, and (auxiliary) produce the teacher’s rationale — so the rationale is a *training target*, not an *inference input*. A 770M T5 thereby beat a 540B PaLM few-shot teacher using 80% of the labelled data, and matched full-data fine-tuning with as little as 12.5% on e-SNLI. This is the conceptual ancestor of internalised reasoning, and the precedent for an explanation-aligned auxiliary head on a discriminative scorer; e-SNLI [22] is the canonical dataset pairing labels with natural-language explanations for exactly this predict-and-explain supervision.

Distillation versus on-small-model RL. R1-Distill [43] curated $\sim 800K$ R1 traces and applied *pure SFT* to open bases (Qwen2.5 1.5B–32B, Llama 8B/70B); the distilled 32B (AIME-2024 72.6 / MATH-500 94.3) far ex-

ceeds RL-from-scratch on the same base (47.0 / 91.6), the cleanest published confirmation of the distill-first verdict. SCoTD [97] shows 125M–1.3B students gain CoT ability when trained on many diverse teacher chains, and Magister et al. [107] established that CoT fine-tuning transfers reasoning to small students.

Label-only rationale bootstrapping. When only task labels exist (the project’s situation — causal-direction labels, no gold rationales), rationales can be self-generated. STaR [193] keeps self-produced rationales that reach the correct answer and *rationalises* the rest by conditioning on the gold answer; rejection-sampling fine-tuning (RFT) [190] is its one-shot best-of- N variant. COLLATE [130] goes further with a teacher-free signal: it ranks candidate rationales by how much each *raises the model’s own likelihood of the gold answer*, forming DPO preference pairs — no judge, no larger teacher, needing only labels, and keeping a sample only when the winning rationale truly raises gold-answer likelihood.

Rationale distillation into a cross-encoder (think-free, zero added latency). The unifying recipe is to add a rationale-derived auxiliary loss to a discriminative head: a generative [rationale] decoding head (Distilling Step-by-Step), or an embedding-alignment head that pulls the classifier’s hidden state toward a (frozen) rationale embedding via cosine/contrastive loss. TFRank’s think-free score [54] is the ranking instantiation. At inference only the label/score head runs, so reasoning improves accuracy at *zero* added latency — exactly what a production DeBERTa-class cross-encoder needs, since it cannot emit `<think>` tokens but can be regularised toward reasoning-relevant features.

6.5 Relevance to causal embedding for AEP/AJO

This section is the backbone of the project’s core thread: *add reasoning to a $< 1B$ causal scorer*. The repository’s strongest ordering model is a DeBERTa-v3-large cross-encoder, which cannot generate reasoning tokens, so genuine reason-then-answer must be *internalised*: the literature converges on TFRank’s think-free pointwise scorer (with a demonstrated 0.6B checkpoint) and Distilling-Step-by-Step / e-SNLI-style auxiliary rationale heads as the way to get reasoning gains at the latency of the current cross-encoder. The training verdict is equally clear: with $\sim 100K$ – $260K$ verifiable labels but no gold rationales, the right pipeline is *bootstrap rationales* (STaR/RFT/COLLATE) $\rightarrow$ *SFT-distil* the behaviour into a sub-1B (or cross-encoder) scorer $\rightarrow$ a short *GRPO/Dr. GRPO* polish with a format reward plus a verifiable correctness reward for the pairwise causal direction and a dense NDCG/Kendall- τ reward for the ordering head — never RL from scratch. The pairwise causal-

precedence judgments this scorer produces are the input to the global-ordering machinery of Sec. 9.

7 Latent and Recursive Reasoning: HRM, TRM, and Iterative Refinement

Chain-of-thought (CoT) reasoning verbalizes intermediate steps as tokens. An alternative family keeps the reasoning chain entirely inside the model’s continuous hidden state—*latent reasoning*—trading interpretability for bandwidth, speed, and the ability to represent superpositions of partial solutions [69, 201]. At the structured-puzzle extreme of this family sit two recent tiny recurrent solvers, the Hierarchical Reasoning Model (HRM) and the Tiny Recursive Model (TRM), which solve constraint-satisfaction tasks such as Sudoku, mazes, and ARC-AGI with 5–27M parameters and no language at all. This section surveys that landscape and connects it to the “think-free” reasoning theme of Section 6, because workflow ordering is itself a small combinatorial puzzle.

7.1 Adaptive computation and recurrence-in-depth

The lineage begins with Adaptive Computation Time [66], which lets a recurrent network predict a per-step halting probability and thus spend a variable number of “ponder” steps per input. Universal Transformers [44] apply the same idea in depth: a single weight-tied Transformer block is iterated, with per-token ACT halting, so effective depth becomes data-dependent. PonderNet [12] reformulates halting as a differentiable probabilistic model, giving unbiased, low-variance gradients. Deep Equilibrium Models (DEQ) [11] push recurrence to its limit: the output is defined as the *fixed point* $z^* = f_\theta(z^*, x)$ of a weight-tied map, found by root-finding, with constant-memory backpropagation through the equilibrium via the implicit function theorem. These works share one principle—*decouple computational depth from parameter count by iterating a small block*—which HRM and TRM inherit directly.

At language-model scale, Geiping et al. [63] train a 3.5B “recurrent-depth” model (Huginn) whose core block can be unrolled for an arbitrary number of steps at test time, scaling test-time compute per token without growing parameters and performing a latent CoT. Coconut [69] feeds the model’s last hidden state back as the next “continuous thought,” and shows that a continuous thought can encode several candidate next steps at once—an implicit breadth-first frontier—solving graph reachability in $O(\text{diameter})$ latent steps where discrete CoT needs $O(n^2)$. Quiet-STaR [194] learns token-level internal rationales that are never shown but improve next-token prediction, bridging latent reasoning and the internalized/think-free training of Section 6.

7.2 HRM: hierarchical recurrent convergence

HRM [170] couples two recurrent modules: a slow high-level module for abstract planning and a fast low-level module for detailed computation. The low-level module iterates to a local equilibrium conditioned on the current high-level state, after which the high-level state updates—“hierarchical convergence”—so a single forward pass performs many internal reasoning steps with no supervision of the intermediate process. HRM uses **deep supervision** (repeated forward segments, each carrying state forward and incurring a loss), a **one-step gradient approximation** that treats the converged inner state as a fixed point and backpropagates roughly one step (in the DEQ spirit) to avoid full backpropagation-through-time, and an **ACT/Q-learning halting** head that decides how many segments to run. With ~ 27 M parameters trained on ~ 1000 examples per task and no pre-training or CoT data, HRM reports near-perfect Sudoku-Extreme and Maze-Hard accuracy and competitive ARC-AGI scores ($\sim 40\%$ on ARC-AGI-1).

What actually drives HRM (an honest caveat). An independent re-implementation and ablation by the ARC Prize team [8] is essential context. They find that the *hierarchical* architecture contributes little: a same-sized ordinary Transformer comes within ~ 5 percentage points. The real driver is the *outer refinement loop*—revising the predicted answer across multiple passes. Going from one pass to two adds ~ 13 points, and one to eight roughly doubles public-evaluation accuracy; moreover, *training* with many refinement steps matters far more than using them only at inference (>15 points even when inference uses a single loop). Data augmentation is necessary but cheap (~ 300 augmentations reach near-maximum). Critically, the setup is *transductive*: a per-puzzle identifier embedding means the model only applies to puzzle IDs seen in training, cross-task transfer is small, and most performance is memorizing solutions for the specific evaluation tasks. The transferable lesson is therefore “*train a recurrent solver with an outer refinement loop and augmentation under deep supervision,*” not the biological hierarchy and not the transductive identifier.

7.3 TRM: less is more

TRM [83] distills HRM to its working core and improves on it. A *single* tiny two-layer network (~ 5 – 7 M parameters) recurses over a triple of an embedded question x , a current answer y , and a latent state z . The inner recursion runs, by default, $n=6$ steps of $z \leftarrow \text{net}(x+y+z)$ followed by one update $y \leftarrow \text{net}(y+z)$; an outer loop runs $T=3$ such blocks ($T-1$ gradient-free to pre-converge the state, then one with gradients), repeated under deep supervision for up to 16 steps with the state detached between steps. TRM *drops* both HRM’s biological two-

timescale story and its fixed-point/implicit-function justification, instead backpropagating through the full recursion; the ablation shows the one-step gradient approximation actually *hurts* (87.4% $\rightarrow$ 56.5% on Sudoku-Extreme). It uses a single binary halting loss (one forward pass instead of HRM’s two) and exponential moving-average weights for stability on scarce data (-7.5 points without it). For small fixed grids it replaces self-attention with an MLP-mixer token mix, cheap when sequence length $L \leq$ width. TRM reports Sudoku-Extreme 87.4% (HRM 55.0%), Maze-Hard 85.3% (HRM 74.5%), ARC-AGI-1 44.6% (HRM 40.3%) and ARC-AGI-2 7.8% (HRM 5.0%)—surpassing much larger LLMs on these grids with under 0.01% of their parameters. The same transductive, augmentation-heavy regime as HRM applies, so the ARC-team critique of the *setup* carries over.

Token CoT versus latent reasoning. Explicit CoT spends decoding budget and is interpretable but brittle and slow; latent reasoning keeps the chain in continuous state, gaining bandwidth and removing decoding cost at the price of opacity [201]. HRM and TRM are the extreme: *no language at all*, only iterative refinement of a structured answer tensor. This is precisely why they fit combinatorial tasks with a structured output—ordering, assignment, constraint satisfaction—better than they fit free-text reasoning.

7.4 Relevance to causal embedding for AEP/AJO

Workflow ordering in our setting is structurally a small permutation / constraint-satisfaction puzzle: given N steps and a calibrated $N \times N$ pairwise precedence tensor from the cross-encoder, find a global order maximally consistent with the confident edges while resolving cycles—exactly the minimum-feedback-arc-set / Kemeny objective of Section 7, and exactly the “make the grid globally consistent” task TRM solves. This suggests a concrete, sub-1B option: a frozen encoder front-end (the winning DeBERTa-v3 cross-encoder, since HRM/TRM operate on tensors, not text) produces step embeddings and the relation tensor, and a tiny (~ 5 – 7 M-parameter) TRM-style recursive head iteratively refines a soft precedence/permutation matrix under deep supervision with a Kendall- τ -aware listwise loss and ACT halting. This is a *learned* alternative or complement to MFAS/Kemeny at the aggregation stage, and a concrete realization of the listwise/global orderer recommended in Sections 7 and 11—able to use step *content*, not just scalar scores, to break ties and to perform soft transitive closure and cycle repair in latent space, directly targeting the 0/30 aggregation collapse.

The honest verdict, however, is bounded. Latent recursion does *not* help the pairwise primitive: a recursive head over frozen, independently-pooled embeddings forfeits the cross-attention that makes the cross-

encoder strong (the same weakness that left the bi-encoder at ~ 0.41 pairwise accuracy and 0/30). It does *not* beat MFAS/Kemeny when the pairwise signal is already clean—on the cross-encoder’s ~ 0.96 signal, plain robust aggregation already orders every evaluated workflow correctly (Section 11) at no training cost. Its value is confined to the *decent-but-noisy, cyclic, partial-order* regime, and it inherits the ARC-team caveats: the architecture matters less than the refinement loop and augmentation, and our scarce supply of gold full-workflow orders (~ 30 in the AJO test, versus the ~ 1000 -examples-plus-augmentation regime these solvers assume) creates a real overfitting/memorization risk. We therefore recommend a TRM-style orderer as a low-cost *Track-B research bet*—a learned listwise ordering head atop the frozen cross-encoder, prototyped and benchmarked head-to-head against calibrated MFAS/Kemeny on the same relation tensors—pursued only after the no-training Tier-1 aggregation fix (Section 11) is in place.

8 Reasoning-Augmented Embeddings under 1B Parameters

Section 6 surveyed reasoning *rerankers*—generative models that read a candidate and emit a score. This section addresses the complementary question: can the *embedder itself* reason? A dense retriever maps text to a vector once and compares with a cheap kernel, so reasoning, if any, must happen *before pooling*. We survey the 2024–2025 line of *reason-then-embed* and reasoning-trained retrievers, the unified generate-and-embed models, and the instruction-controllable embedders, all under the project’s < 1 B-parameter budget, and assess how—and whether—to add reasoning to a Qwen3-Embedding-0.6B backbone for directional causal retrieval.

8.1 Why plain embedders fail on reasoning

The motivating evidence is the BRIGHT benchmark [159], the first retrieval benchmark whose relevant documents require multi-step reasoning rather than surface matching. Strong models that reach ~ 59 nDCG@10 on standard benchmarks collapse to ~ 18.3 on BRIGHT, with a best score of only ~ 24.3 ; crucially, supplying chain-of-thought (CoT) reasoning to expand the query improves retrieval by up to $+12.2$ nDCG@10. This is the embedding-side analogue of the reranker finding: a single similarity computation cannot perform the inference that connects a query to a non-obviously-relevant document, but an explicit reasoning step can bridge the gap. Our project’s ordering task is exactly such a case—two workflow steps may be lexically and topically similar yet stand in a non-obvious *causal precedence* relation that no symmetric similarity recovers (Sec. 3).

8.2 Reason-then-embed

Test-time reasoning (training-free). The cheapest form generates auxiliary text at query time and embeds *that*. HyDE [61] zero-shot prompts an instruction LLM to write a *hypothetical document* answering the query, then embeds the hypothetical with an unsupervised encoder; query2doc [172] few-shot generates a pseudo-document and concatenates it to the query, lifting BM25 by 3–15% and also helping dense retrievers. Both are training-free “reason-then-embed” at the prompt level, at the cost of one LLM generation per query.

Learned reason-then-embed. O1-Embedder [185] makes the reasoning intrinsic: the model first *generates “thoughts”* for the input query, then embeds the thought-augmented query for dense retrieval. It is trained by jointly (i) behavior-cloning thoughts whose training signal is synthesized by an LLM-expert and refined by a retrieval committee, and (ii) standard contrastive retrieval. The decisive deployment property is that reasoning is applied *query-side only*: documents are embedded normally, so the corpus index remains precomputable. ReasonIR [154] instead targets the *training data*: its ReasonIR-Synthesizer fabricates varied-length reasoning queries paired with hard, plausibly-related-but-unhelpful negatives, reaching 29.9 nDCG@10 on BRIGHT without a reranker (8B, above our budget, but the recipe transfers to a small student). RaDeR [41] mines reasoning supervision from Monte-Carlo-Tree-Search trajectories over mathematical problem solving and is the first dense retriever to beat BM25 when queries are CoT steps; DIVER [104] wraps a reasoning-fine-tuned retriever in a four-stage pipeline (document cleaning, iterative LLM query expansion, retrieval, pointwise reranking) for ~41.6 nDCG@10 on BRIGHT. The common thread—reasoning helps embedding retrieval, and it is most naturally and cheaply applied to the query.

8.3 Unified generate-and-embed and decoder-LLM embedders

A reason-then-embed system needs a backbone that can both generate and embed. GritLM [119] shows one model can do both via *instruction routing*—generation under a generative instruction, embedding (mean-pooled, bidirectional attention) under an embedding instruction—at no loss to either task, and it speeds retrieval-augmented generation by reusing the same forward states. This is the blueprint for “one Qwen3-0.6B in two modes.” Decoder-only LLMs are turned into embedders by LLM2Vec [16] (enable bidirectional attention, adapt with masked-next-token prediction, then SimCSE-style contrastive training), by echo embeddings [157] (repeat the input and pool the second copy so early tokens attend to later ones, a training-free fix for causal-attention pooling), and earlier by SGPT [116]. Qwen3-Embedding [199] is the pro-

duction realisation under our budget: a 0.6B decoder-LLM embedder with last-token pooling, instruction-aware queries, 32k context, and Matryoshka dimensions (32–1024), trained by large-scale synthetic weak supervision, supervised contrastive fine-tuning, and model merging, with a matching generative Qwen3-Reranker-0.6B. Because its backbone is a full causal LM, the generative capacity needed for reason-then-embed is already latent in the weights.

8.4 Instruction-controllable (directional) embeddings

The other lever is instruction conditioning. INSTRUCTOR-style task prefixes and Promptriever [177]—the first zero-shot *promptable* retriever, instruction-trained on an MS MARCO-derived set and gaining +14.3 p-MRR on FollowIR—show that a free-text instruction can steer what an embedder treats as relevant. The benchmarks temper this: INSTRUCTIR [129] and FollowIR [176] find instruction-following in retrieval real but fragile, with some instruction-tuned retrievers *underperforming* their bases. For our problem the relevance is direct: “represent this step *as a cause*” versus “*as an effect*” is an instruction, and embedding the same text under two roles yields an *asymmetric* comparison that plain cosine— $\text{sim}(a, b) = \text{sim}(b, a)$ —cannot express. This complements the asymmetric-geometry treatment of Sec. 3: instruction conditioning is a learned, prompt-level route to directionality, an order/box/cone head is the geometric route, and a learned cause/effect projection $s(A \rightarrow B) = \langle W_c \mathbf{e}_A, W_e \mathbf{e}_B \rangle$ sits between them.

8.5 Architectures for adding reasoning to Qwen3-0.6B

We enumerate concrete options and their trade-offs for the AEP/AJO tasks. **(A) Reason-then-embed** (O1-Embedder style): Qwen generates a short causal “thought” for the query, then embeds query+thought. Pro: genuine reasoning, BRIGHT evidence. Con: generation latency—acceptable *only query-side*, since generating a thought per corpus document would destroy the bi-encoder’s precompute/caching. **(B) Unified GritLM-style**: one Qwen3-0.6B emits a rationale and embeds via instruction switching—flexible but requires joint training, and the project’s own GritLM-7B probe stayed semantic and ignored a causal instruction zero-shot (Sec. 3). **(C) Think-free / distilled**: train with rationales, use none at inference—the latency-safe embedder route, identical in spirit to TFRank’s think-mode switch (Sec. 6); preferred when the embedder must run at index speed. **(D) Latent reasoning before pooling**: looped/recurrent depth or learned “think” tokens consumed before last-token pooling—no token generation, but speculative for embedders (cf. the recurrent-depth discussion). **(E) Gen-**

erative pointwise reasoner-reranker: Qwen3-0.6B reasons then scores a pair’s precedence—the most accurate directional option, but a reranker not an embedder, treated in Sec. 6. **(F) Asymmetric/instructed directional embedder:** cause/effect instructions plus a learned asymmetric scoring head—directional at embedding speed and fully precomputable, at a lower accuracy ceiling than cross-attention.

8.6 Relevance to causal embedding for AEP/AJO

For the three project tasks the fit is specific. *Pairwise causal direction* is inherently antisymmetric, so the highest accuracy comes from a cross-attentive reasoner (option E, Sec. 6) and the embedding-speed alternative is the asymmetric head (F). *Follow-up reranking*—“given this step, what should one ask or do next?”—is a reasoning-intensive, BRIGHT-like query and is the natural home of reasoning-augmented embedding: query-side reason-then-embed (A) over Qwen3-Embedding-0.6B with an asymmetric head (F), with instruction prompting (HyDE/query2doc/Promptriever) as a zero-training first probe. *Workflow ordering*, by contrast, is shown in Sec. 3 to be a *signal-plus-aggregation* problem that the DeBERTa cross-encoder already solves (21/30 versus 0/30, made aggregable by robust MFAS/Kemeny ordering): an embedder, reasoning or not, is at best a stage-1 directional recaller here. The verdict is therefore that adding reasoning to Qwen3-0.6B is worth it as a *complement*—query-side reason-then-embed plus an asymmetric cause/effect head for follow-up reranking and directional recall—and that the latency-disciplined form keeps the corpus side precomputable, but it should *not* replace the cross-encoder-plus-aggregation ordering pipeline. The SOTA-grounded expectation under 1B is a moderate gain on reasoning-intensive retrieval, contingent on the O1-Embedder/ReasonIR synthetic-data recipe, not a breakthrough on strict ordering.

9 From Pairwise Scores to a Global Order: Sentence Ordering and Rank Aggregation

The reasoning scorer of Sec. 6 emits, for the steps of a workflow, a matrix of pairwise directional scores $P[i, j] = P(\text{step}_i \text{ causally precedes step}_j)$. The remaining problem is *aggregation*: turning these noisy, possibly cyclic, possibly partially-ordered judgments into a global order. This is the locus of the project’s headline failure — a naive tournament/topological sort scores 0/30 on the AJO ordering test while a stronger pairwise model reaches 21/30, a gap whose signature is a brittle *aggregation* step, not a weak pairwise model. This section surveys the ordering task, why topological sort fails, the robust rank-aggregation alternatives, calibration and partial-order handling, and

listwise objectives.

9.1 The sentence/step-ordering task

The NLP *sentence-ordering* task (recover the order of shuffled sentences) is the closest analogue to workflow-step ordering and has directly studied “pairwise-then-assemble” against end-to-end models.

Pairwise-then-topological-sort (B-TSort). Prabh-moye et al. [135] reframe ordering as constraint solving: a BERT classifier predicts pairwise relative-order constraints, then a *topological sort* of the constraint graph produces the order (TSort / B-TSort). This is essentially the project’s current pipeline. Earlier, Chen et al. [31] combined a pairwise model with *beam search* to assemble the order.

Listwise / pointer orderers. To avoid committing to independent pairwise decisions, pointer/seq2seq orderers encode all sentences jointly and *decode* the permutation. Gong et al. [64] and Logeswaran et al. [103] use pointer networks explicitly to “alleviate error propagation”; BERSON [36] adds a BERT-based relational pointer decoder; RankTxNet [91] scores each sentence and sorts, supporting pointwise/pairwise/listwise losses in one architecture. As re-tabulated by Re-BART [14], the pointer decoder of BERSON reports markedly higher τ and PMR than B-TSort (e.g. on NIPS, PMR 48.0 vs 32.6), consistent with the error-compounding thesis below.

Metrics. Standard measures are *Kendall’s τ* (rank correlation), *Perfect Match Ratio* (PMR, the fraction of instances ordered *entirely* correctly — the strict analogue of the project’s 0/30 metric), and longest-common-subsequence (LCS) overlap with the gold order.

9.2 Why naive topological sort fails on noisy scores

A topological sort *requires a DAG*: on *any* directed cycle it cannot produce an order at all, and forcing acyclicity by dropping arbitrary edges scrambles the result. Noisy pairwise causal scores are almost always cyclic ($A \rightarrow B \rightarrow C \rightarrow A$), so topological sort is simply inapplicable — the most likely cause of the 0/30 result. The deeper issue is *error compounding*: a pairwise-then-assemble pipeline makes $O(n^2)$ independent binary decisions, and a single high-confidence wrong edge can reorder an entire chain. If per-pair accuracy is a , exact-sequence success is closer to $a^{\# \text{critical pairs}}$ than to a — precisely how a 21/30 per-pair-strong model can collapse to 0/30 exact-order. A robust global objective, rather than greedy commitment, is required.

9.3 Rank aggregation from pairwise comparisons

The principled fix replaces topological sort with an aggregator that consumes the *weighted* pairwise data and degrades gracefully under noise.

Probabilistic models: Bradley–Terry / Plackett–Luce. Bradley–Terry (BT) [20] gives each item a latent worth β_i with $\Pr(i \succ j) = \sigma(\beta_i - \beta_j)$ — logistic regression on score differences, the family Elo belongs to. Plackett–Luce (PL) generalises BT from pairs to full rankings via a chain of choose-best softmaxes. Maximum-likelihood inference of any Luce-axiom model is the stationary distribution of a Markov chain; Luce Spectral Ranking (LSR) approximates it in one spectral step and I-LSR converges to the MLE [109]. The `choix` library [108] implements the pairwise variants (`ilsr_pairwise`, `mm_pairwise`, `opt_pairwise`, `rank_centrality`). BT/PL are parametric: a true Condorcet cycle is fit to the closest *transitive* score vector (so cycles are absorbed, not represented), and the MLE is unique only when the comparison graph is strongly connected.

Spectral and win-counting baselines. Rank Centrality [120] scores items by the stationary distribution of a random walk that passes mass from losers to winners; it is minimax-optimal for top- K recovery under standard noise [32] and costs one power iteration per step. Plain *win-counting* (Borda/Copeland) is provably optimal, robust and efficient *with no parametric assumption* [153] — the most model-robust baseline and an excellent initialiser. HodgeRank [80] decomposes the pairwise flow (via a single sparse graph-Laplacian solve) into a *gradient* (rankable) part plus *curl* and *harmonic* (cyclic, inconsistent) parts, returning a global ranking *and* a certificate of how rankable the data is — large cyclic energy signals that no clean chain exists.

Minimum Feedback Arc Set = Kemeny / minimum-violations ranking. The principled global objective is the *minimum-weight feedback arc set* (MFAS): the linear order minimising the number (weight) of backward arcs — equivalently the order disagreeing with the fewest pairwise edges. On the majority weighted tournament this is exactly the *Kemeny consensus*, the order minimising total Kendall- τ distance and the maximum-likelihood ranking under Condorcet-style noise — a justification topological sort lacks. It *degrades gracefully*: on a true DAG it reproduces the topological order with zero backward arcs; on cyclic input it returns the best-supported order. MFAS is NP-hard, even on tournaments [5, 84], and Kemeny aggregation is NP-hard with as few as four input rankings [13, 48]; but tournaments admit a PTAS [89] and excellent practical

heuristics. The standard heuristic is Eades–Lin–Smyth greedy [49] (peel off sinks and sources, then pick the vertex maximising outdeg – indeg; linear time) followed by local search (swaps/reinsertions reducing backward-arc weight); for $n \lesssim 30$ –40 an exact ILP (order variables x_{ij} , triangle constraints $x_{ij} + x_{jk} + x_{ki} \leq 2$, minimise backward weight) is tractable. Note a practical pitfall: **NetworkX has no `feedback_arc_set`**; it supplies only building blocks (`topological_sort`, `topological_generations`, `simple_cycles`). The library that provides FAS is **igraph** (`feedback_arc_set` with `approx_eades/exact_ip`) [35].

9.4 Calibration and confidence-weighted aggregation

Aggregators consume $P[i, j]$ as a *weight*, not just a sign, so miscalibration directly distorts the global order: an overconfident wrong edge can dominate a weighted-MFAS objective and flip the result. The cheap, standard fix is *temperature scaling* [68] — a single scalar T , with calibrated probability $\sigma(z/T)$, fit to minimise validation NLL; because it scales all logits equally it leaves the arg-max (and thus accuracy) unchanged while fixing confidences. Platt scaling [134] and isotonic regression [191] are the parametric and non-parametric alternatives. Calibrated scores then justify *confidence-weighted edges*: set the arc weight to the log-odds $w_{ij} = \log(P[i, j]/(1 - P[i, j]))$ (the natural BT/weighted-FAS weight) or the margin $P[i, j] - 0.5$, so a low-confidence wrong edge costs little and cannot flip the order, while only calibrated P makes a threshold τ correspond to a fixed reliability level.

Abstention and partial orders. A workflow is generally a *partial* order, not a chain: genuinely parallel steps form an *antichain* and must be left unordered. Rather than force every pair into a direction, *abstain* (emit NEUTRAL) when $P[i, j] \approx 0.5$ within a calibrated band, keep only high-confidence oriented edges, and output a *DAG*. `topological_generations` then stratifies the DAG into layers that are exactly the parallel-step antichains; Dilworth’s theorem [46] relates the largest antichain (maximum parallelism) to the minimum chain decomposition. Forcing a total order over parallel steps both misrepresents the workflow and manufactures spurious “errors.”

9.5 Listwise losses and partial-order evaluation

Listwise losses. If the orderer can be trained end-to-end, a listwise objective optimises the whole permutation jointly and avoids the $O(n^2)$ -independent-decision compounding. ListNet [23] defines a PL distribution over permutations and minimises cross-entropy to the gold (top-one) distribution; ListMLE [180] is the negative log-likelihood of the gold permutation under Plackett–Luce (i.e. ListMLE = PL NLL), and is convex and consistent.

An autoregressive pointer decoder implements the *same* PL factorisation operationally (each step a softmax over remaining items). RankTxNet’s controlled comparison [91] isolates the objective: listwise (ListMLE) beats pairwise beats pointwise on most datasets, matching the error-compounding analysis of Sec. 9.2.

Partial-order evaluation. Exact-chain matching (PMR) wrongly penalises a correct-but-different linear extension of parallel steps, so evaluation should be tie-aware. Use Kendall’s τ_b/τ_c (tie-corrected denominators) [88], Goodman–Kruskal γ [65], and the Fagin et al. partial-ranking τ -distance with a tie penalty $p \in [0, 1]$ ($p = \frac{1}{2}$ neutral) [52, 53]. The most interpretable scheme is *precedence precision/recall/F1*: represent the gold partial order as its set of precedence constraints $\{(i, j) : i \text{ must precede } j\}$ and score the prediction as pairwise classification, crediting any correct precedence regardless of which valid linear extension was chosen and not penalising the ordering of genuinely parallel pairs.

9.6 Relevance to causal embedding for AEP/AJO — the fix for 0/30

This section is *the* fix for the 0/30 ordering failure. The diagnosis is structural: topological sort cannot survive the cycles that noisy pairwise causal scores inevitably produce (Sec. 9.2), and exact-chain scoring punishes valid alternative orderings of parallel steps. The minimal-change, maximal-impact pipeline keeps the existing pairwise reasoning reranker of Sec. 6 and replaces only the aggregation: (1) *calibrate* the pairwise scores with temperature scaling; (2) build a *weighted tournament* with log-odds edge weights (do not threshold to a hard DAG); (3) aggregate with *weighted MFAS / Kemeny* — initialise by win-counting (Copeland/Borda), refine with Eades–Lin–Smyth greedy plus local search (or an exact ILP for the short workflows here, or igrap’s `feedback_arc_set`; not NetworkX, which lacks it); optionally run `choix` Rank Centrality and HodgeRank in parallel to obtain scores and a cyclic-inconsistency certificate; (4) *abstain* on near-0.5 pairs and emit a DAG, recovering parallel steps as antichains; (5) *evaluate as a partial order* with tie-aware τ_b and precedence precision/recall. Longer term, a listwise (ListMLE) objective or a pointer decoder trained end-to-end optimises full-sequence accuracy directly and removes per-pair error compounding. Steps (1)–(3) plus (5) together address the precise reasons topological sort scores 0/30.

10 Follow-up Question Ranking and Conversational Retrieval

A causal embedding model for procedural documentation is not consumed in isolation: in the AEP/AJO assistant it sits behind a conversational surface that must decide

what the user should ask next. This section surveys the research landscape underpinning that capability—follow-up reranking, clarifying-question generation and ranking, multi-turn conversational retrieval, and proactive/mixed-initiative dialogue—and argues that ordering follow-ups by *causal/next-step progression* rather than by mere semantic relevance is the distinguishing requirement for our setting.

10.1 The follow-up reranking task in this project

The concrete task instance in the repository is a *slate-reranking* problem. The production data file (`prod_jan_feb_26-aep-ajo_follow-up-queries.json`) is a list of 334 records, each of the form

```
{"question": <user query>,
 "follow_up_queries": [[ fq1, ..., fq10 ]]}
```

i.e. a single user question paired with a slate of exactly ten candidate follow-up questions (the inner list is wrapped once). For the anchor “*What is Journey Pause feature*” the candidates range over permissions, the hold-vs-discard semantics at action nodes, scheduled audience runs during a pause, organization-wide held-profile limits, and re-entrance/timeout interaction—a mix of *prerequisite*, *consequence*, and *lateral* questions. The reranking harness (`rerank_followup.py`) flattens every record into (`question`, `follow_up`) pairs, scores each pair with the fine-tuned `DirectionalClassifier` bi-encoder (score = $\sigma(\text{MLP}([a; b; a - b; a \cdot b]))$, the architecture from Section 3), and sorts the slate descending by score. Crucially the model used is the *directional precedence* classifier, so the score approximates $P(\text{question causally precedes/enables follow-up})$ rather than topical similarity—this is exactly the “rank by logical progression” signal we want, and it is what differentiates the task from a generic relevance reranker. The slate is fixed and there is no gold ordering in the file, so evaluation must come from a downstream signal (click logs, LLM-as-judge ordering, or a held-out human ranking).

10.2 Clarifying-question generation and ranking

When a user query is under-specified, a proactive system can either retrieve or *ask*. Rao and Daumé III [142] framed clarifying-question *ranking* as maximizing the expected value of perfect information (EVPI) over StackExchange post–comment data, selecting the question whose likely answer most changes the system’s belief. Aliannejadi et al. [3] introduced the offline **Qulac** methodology and the influential SIGIR’19 formulation of *asking clarifying questions in open-domain information-seeking conversations*, later scaled into the **ClariQ**/ConvAI3 challenge [4] with explicit *clarification-needed* labels (the `clariq` TSV in our repo carries `topic_id`, `initial_request`,

facet_desc, question, answer columns and a 4-level clarification-need score). **ClarQ** [92] bootstrapped $\sim 2\text{M}$ clarification questions across 173 StackExchange domains for generation, and **CLAMBER** [197] contributes an LLM-oriented taxonomy of $\sim 12\text{K}$ ambiguous queries, finding that current LLMs are still weak at both *recognizing* ambiguity and *generating* a useful clarifying question. At web scale, **MIMICS** [192] mined real search clarification panes (the “People also ask”/clarification-pane lineage). These threads share our core sub-problem: given a query, produce and *rank* a small slate of next questions; the AEP follow-up slate is the documentation-grounded, ten-candidate version of this.

10.3 Conversational and multi-turn retrieval

Follow-ups live inside multi-turn sessions, where the central difficulty is *context-dependence*: a follow-up like “what permissions does it need?” is meaningless without its antecedent. Two solution families dominate.

Query rewriting. **CANARD** [50] reformulates context-dependent questions into self-contained ones; T5-based rewriters (**T5QR**, 100) became the de-facto sequence-to-sequence approach, turning a conversation into a standalone query that any ad-hoc retriever can consume. **QReCC** [6] packages 14K conversations as an end-to-end “rewrite→retrieve→answer” benchmark (qrecc_train.json/qrecc_test.json in the repo).

Conversational dense retrieval. Rather than rewrite, **ConvDR** [189] trains a session encoder to embed the full multi-turn context directly, distilling from the embedding of an oracle rewrite (teacher–student) so it works few-shot. **TopiOCQA** [1] stresses *topic switching*: 3,920 conversations averaging 13 turns over ~ 4 documents each, forcing retrieval that detects when the user pivots—directly analogous to a user jumping from “pause a journey” to “read-audience scheduling.” Most recently, **MTRAG** [87] is a human-generated multi-turn RAG benchmark (110 conversations, ~ 7.7 turns, four domains incl. Cloud) explicitly probing non-standalone and unanswerable later-turn questions; the repo’s mtrag/conversations.json mirrors this format with per-turn retriever traces.

10.4 Proactive and mixed-initiative agents

Reranking a fixed slate is a special case of *proactivity*: deciding, unprompted, what to surface next. Deng et al. [45] survey proactive dialogue (clarification, asking, recommending, non-collaborative initiative). **ProactiveAgent** [105] (the proactive_agent dir) trains an agent on $\sim 6.8\text{K}$ real events to *anticipate* the user’s next task and offer assistance before being asked, with a reward

model scoring whether an intervention was wanted. On the generation side, **FollowupQG** [111] (the followupqg split: {question, answer, follow-up, relation} tuples from Reddit ELI5) targets *information-seeking* follow-up generation, showing that model follow-ups remain markedly less informative than human ones—evidence that good follow-ups require genuine reasoning about what new knowledge the answer leaves open, not surface paraphrase.

10.5 Reranking methods and the causal/next-step signal

The reranker toolkit spans pointwise cross-encoders (**monoBERT**, 126; **monoT5**, 128) that jointly attend over (query, candidate), and listwise LLM rerankers (**RankGPT**, 160; distilled into the 7B **RankZephyr**, 136) that emit a full permutation “[3] > [1] > ...” over the slate. All of these optimize *relevance*. The thesis of this survey is that a follow-up slate should instead be ordered by *logical/causal progression*: the next question a user *should* ask is typically the immediate prerequisite or the immediate consequence of the current step, not merely the most topically similar one. Two ingredients combine cleanly: (i) the joint cross-attention of a cross-encoder/LLM reranker, which Section 3 and the repo’s ordering experiments show generalizes better to non-adjacent pairs; and (ii) a *directional precedence* score $s(q, fq) \approx P(q \prec fq)$ from the causal embedding model. A listwise reranker can be supervised or prompted to sort by s rather than by relevance, and the same pairwise scores feed the rank aggregator of Section 11 when a globally consistent ordering of the slate is required.

Relevance to causal embedding for AEP/AJO.

The follow-up slate is the most direct deployment of the causal embedding primitive: **rerank_followup.py** already scores (*question, follow-up*) with the directional bi-encoder and sorts by it. The literature surveyed here motivates three concrete upgrades for our $< 1\text{B}$ -parameter model. First, treat the slate as a *partial order* of prerequisite/consequence/lateral follow-ups (mirroring CLAMBER’s ambiguity types and proScript’s partial-order scripts in Section 11) rather than a single relevance ranking, so parallel “lateral” follow-ups are not spuriously forced above causal next-steps. Second, adopt a cross-encoder or distilled listwise reranker (RankZephyr-style) so the ordering signal benefits from joint attention—consistent with the repo finding that the DeBERTa cross-encoder dominates on *ordering* despite trailing on flat classification. Third, because follow-ups arrive mid-conversation, a ConvDR/CANARD-style rewrite step should canonicalize the anchor question before causal scoring, so context-dependent follow-ups (“how does it interact with re-entrance?”) resolve to their AEP/AJO referents before precedence is estimated.

11 Procedural and Workflow Understanding

The third target task—*sort the steps of a workflow into ascending order*—is an instance of procedural understanding, a problem the NLP community has studied under several guises: tracking state through a process, inferring goal-step and step-step relations, inducing event schemas, and planning. This section surveys those threads and connects them to ordering the AEP/AJO tutorial steps from documentation and video transcripts.

11.1 Procedural text understanding

The canonical benchmark is **ProPara** [38]: 488 human-authored science procedures with dense per-step annotations of entity *existence* and *location*, where the model must track state changes whose causes are often implicit and must be inferred—the same implicit-causality challenge our pairwise primitive faces. **OpenPI** [163] extends entity-state tracking to open-domain wikiHow procedures. **RecipeQA** [184] (~20K recipes, ~36K QA pairs) includes an explicit *ordering* subtask—reorder shuffled steps—that is structurally identical to our workflow-ordering goal, plus a coherence/cloze subtask. The wikiHow line of work is most directly relevant: Zhang et al. [196] define *Goal-Step Inference* (“learn poses” is a step of “do yoga”), *step inference*, and a *temporal ordering* task (“buy a yoga mat” precedes “learn poses”), reporting a 10–20% gap between transformer models and humans; **VGSI** [186] adds a visual variant over 772K step images. These step-step temporal relations are precisely the forward(1)/reverse(0) precedence labels the repo derives from positional order within a document.

11.2 Script and event-schema induction

Ordering steps is a modern form of *script* learning. Chambers and Jurafsky [27] introduced unsupervised *narrative event chains* and the *narrative cloze* evaluation—predict the held-out event in a partially ordered chain—which directly anticipates next-step prediction over a workflow. **ROCStories** [114] supplies a story-cloze/temporal-ordering benchmark over everyday event sequences. Commonsense event schemas were operationalized by **ATOMIC** [150] (if-then social/physical relations incl. *xNeed* prerequisites and *xEffect* consequences) and made generative by **COMET** [18], which fine-tunes a transformer to emit commonsense tails—a template for emitting prerequisite/consequence edges between AEP steps. **proScript** [146] is the most aligned: it generates *partially ordered* scripts and defines an *edge-prediction* task (given a scenario and unordered events, recover the partial-order DAG; F1= 75.7). Its partial-order stance directly supports the repo’s diagnosis that forcing a *total* order over parallel workflow steps is a root cause of the 0/30 ordering failure.

11.3 Workflow and process modeling with LLMs

Recent work scales procedural understanding to executable workflows. **WorkflowLLM** [55] builds *WorkflowBench*—106,763 orchestration samples over 1,503 APIs from 83 applications—and fine-tunes *WorkflowLlama* to generate API workflows, outperforming GPT-4o with in-context learning and generalizing to out-of-distribution tasks (F1 plan 77.5% on T-Eval). The repo’s `workflowllm_aep_merged_2` dataset (210K pairs) adapts this corpus to AEP. **FlowBench** [181] benchmarks *workflow-guided* agent planning across 6 domains, 22 roles, and 51 scenarios, comparing workflow-knowledge formats (natural language, symbolic code, flowchart) and finding even GPT-4o reaches only ~40–43% success—a flowchart/DAG representation gave the best trade-off, reinforcing a graph-structured rather than linear view of process knowledge. This connects to classical *process extraction* from documents and to hierarchical task networks (HTNs) / task graphs, where a procedure is a partial order of subtasks rather than a flat chain—the representation our rank-aggregation stage should target.

11.4 Procedural planning with LLMs

The planning literature studies *generating* an ordered plan rather than sorting a given one, but the ordering competence transfers. Huang et al. [77] show large LMs decompose high-level goals into mid-level step sequences zero-shot, though raw outputs need grounding to admissible actions; **SayCan** [2] grounds such plans via affordance scoring. **Chain-of-thought** prompting [175] improves multi-step reasoning generally and underlies the rationale-before-verdict strategy proposed for our reranker. Critically, **CaT-Bench** [106] benchmarks LM understanding of *causal and temporal* dependencies in plans (“can step *j* happen before step *i*?”), finding models conflate temporal sequence with causal necessity—exactly the conflation the repo flags between *narration order* and *causal precedence*.

Relevance to causal embedding for AEP/AJO. The downstream goal is to order the 30 three-step AJO orchestrated workflows (`ajo_orchestrated_workflows_flat.json`, fields `t1/t2/t3`) and to recover step order from the 142 AEP video transcripts (`aep_transcripts.json`), where narration order is a noisy proxy for causal order. The surveyed literature suggests four moves, consistent with the repo’s own root-cause analysis. (1) Adopt a *partial-order/DAG* target (proScript, FlowBench’s flowchart format) instead of a forced total order, with a NEUTRAL/abstain class for parallel steps such as the AJO welcome-flow “pause 24h” that is independent of audience enrichment. (2) Frame pairwise scoring as prerequisite/consequence edge prediction (ATOMIC

$xNeed/xEffect$, COMET) so the embedding distinguishes causal necessity from mere sequence, the distinction CaT-Bench shows models miss. (3) Train on the *transitive closure* of within-document orders (not just adjacent pairs) to close the distribution shift the repo identifies between $gap \leq 2$ training and all-pairs evaluation. (4) Aggregate the (possibly cyclic) pairwise scores with a robust ranker—minimum-feedback-arc-set / Bradley–Terry—feeding `order_events.py`, rather than a naive topological sort that lets per-pair errors compound into the current 0/30 result.

12 Data Characterization: AEP/AJO Documentation and Workflows

This section empirically characterizes the corpora that underpin the causal-embedding effort for Adobe Experience Platform (AEP) and Adobe Journey Optimizer (AJO). All statistics below are computed directly from the repository’s JSON/JSONL assets by the reproducible script `implementation/analyze_data.py`; the figures it emits are included verbatim. We deliberately treat the data as small structured files (parsed with `json/csv`) and compute everything with simple counting, so the numbers are auditable and the analysis is decoupled from any model. We first describe the raw sources (documentation pages and video tutorials), then the derived pairwise causal-classification datasets and their tier system, then the labeling procedure (adjacency and `MAX_GAP` windowing) and the distribution shift it induces, and finally the workflow-ordering and follow-up-reranking evaluation sets. We close with implications for modeling.

12.1 Source corpora

Documentation. The pairwise datasets are mined from Adobe Experience League documentation. The deterministic phase-level extractor consumes a snapshot of 3,853 scraped documentation pages (title/URL/content); after markdown, code-fence, table, and callout cleaning, pages are split on H2/H3 headings into action-bearing *phase* paragraphs of 24–56 words. Pages yielding between 3 and 12 phases are retained, producing 1,301 phase-structured workflows. A parallel sentence-level builder restricts to `/tutorials/` pages and keeps only *action* sentences (5–60 words containing an imperative verb such as *select*, *click*, *navigate*, *create*, *configure*), dropping narration markers.

Video tutorials. The transcript corpus (`scrap_transcript/aep_transcripts.json`) contains 142 AEP/AJO video tutorials, each a (`title`, `url`, `toc_path`, `video_id`, `transcript`) record. Transcripts are long and highly variable: the median length is ≈ 792 words, the mean $\approx 1,135$ words, ranging from

Table 3: Primary pairwise causal-classification datasets. Sizes are exact line counts; “% label 1” is the fraction of *precedes* pairs in train; “med. len.” is the median word count of `text_1`; “tier” marks datasets carrying tier meta-data. Every dataset is essentially perfectly balanced by construction (forward/reverse pairs).

Dataset	Train	Val	Test	% lbl1	med. len.	ti
<code>aep_causal_cls34</code>	110,775	6,004	6,004	50.3	103	v
<code>hard_neg_sem</code>	261,210	14,736	6,004	50.3	103	v
<code>ajo_doc_not_t1</code>	88,000	11,200	11,200	50.0	208	v
<code>new_aep_wf_scrap</code>	13,904	1,676	1,548	50.0	44	
<code>workflowllm_m2</code>	210,775	6,004	6,004	50.0	37	
<code>ajo_newstyle</code>	20,897	2,612	2,613	50.0	95	

175 to 9,608 words (Figure 1, left). All videos sit under the `platform-learn/tutorials` tree; by topic area the largest buckets are *destinations* (17), *ai-assistant* (14), *schemas* (13), *privacy* (11), *intro-to-platform* (10), and *sources* (10) (Figure 1, right). The heavy right tail matters operationally: a single transcript routinely exceeds a transformer’s 512-token budget, which is exactly the “text-starvation” failure mode reported for cross-encoders on this data.

12.2 Pairwise causal-classification datasets

The central training signal is a *directional pairwise* format: each row carries `text_1`, `text_2`, and a binary `label`, where `label = 1` means *text_1 causally precedes / enables text_2*. The richer, tiered variants additionally carry `tier_1`, `tier_2` (a curriculum-level index), `sub_1`, `sub_2` (sub-chapter codes), and source URLs. We survey the six datasets referenced as primary in the project context; their split sizes and label balance are shown in Table 3 and Figures 2–3.

Label balance. Because every dataset is built by emitting each ordered pair in both directions (forward $\rightarrow 1$, reverse $\rightarrow 0$), the label distribution is essentially exactly 50/50 in every split (Figure 3). This is a deliberate and important property: it removes the class-prior shortcut, forcing a model to learn *directional asymmetry* rather than a topical or frequency cue. It also means accuracy is directly interpretable against a 50% random floor, and that the genuine difficulty lives entirely in the *direction*, not in the marginal.

Scale and granularity. Sizes span two orders of magnitude, from the $\sim 17k$ -pair `new_aep_wf_scrap` to the $\sim 282k$ -pair `hard_neg_semantic` (Figure 2). Text granularity varies just as widely: the workflow/phase datasets are short (37–44 word medians, single action steps), whereas the conceptual document datasets are paragraph-scale (103–208 word medians). The `hard_neg_semantic`

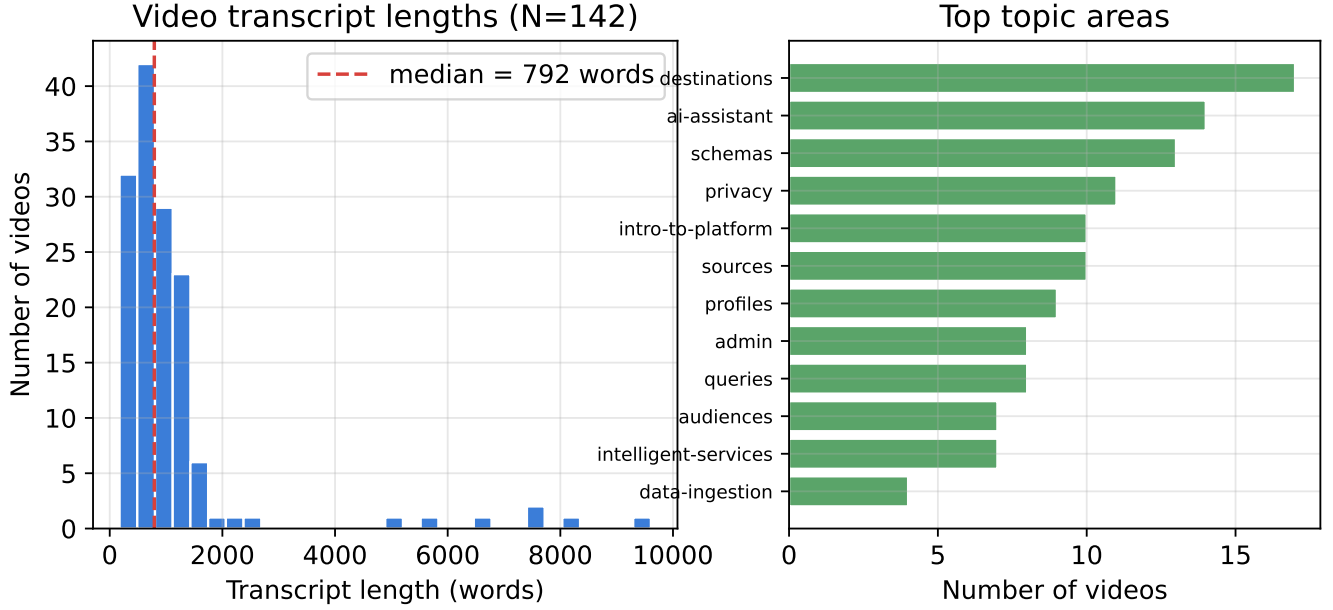


Figure 1: Video-tutorial transcript corpus ($N = 142$). Left: distribution of transcript length in words (median ≈ 792 , long right tail to ~ 9.6 k words). Right: the twelve largest documentation topic areas. The corpus is AEP/AJO-centric and long-form, frequently exceeding standard 512-token encoder budgets.

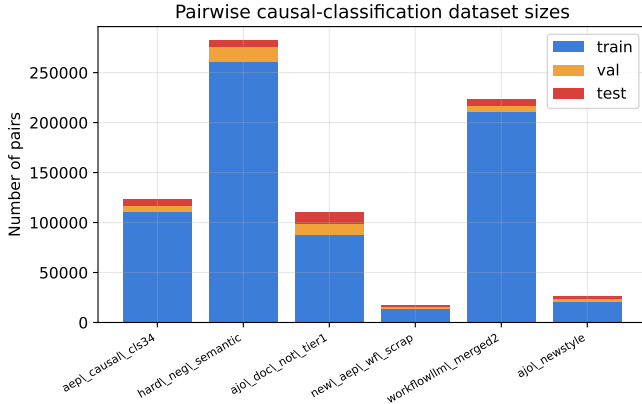


Figure 2: Train/val/test sizes of the six primary pairwise datasets (stacked, exact line counts). Scale ranges from ~ 17 k to ~ 282 k pairs.

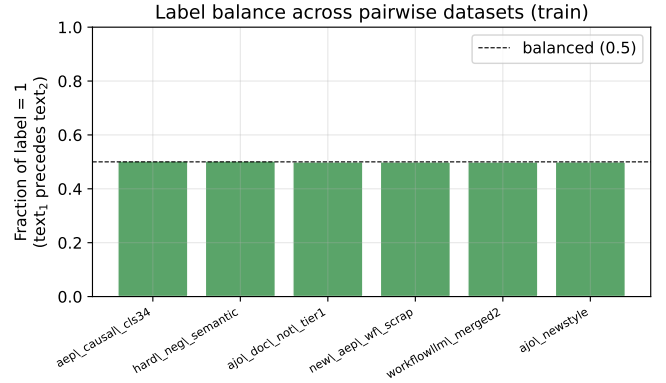


Figure 3: Label balance (fraction of `label = 1` “precedes” pairs, train split). Forward/reverse pair construction yields a $\approx 50/50$ split everywhere, eliminating the class-prior shortcut.

variant augments `aep_causal_cls34` with semantically similar but causally unrelated distractors (an explicit `hard_neg` flag), roughly $2.4\times$ the base training size, to push the model past lexical-overlap heuristics toward genuine causal discrimination.

12.3 The tier system and the cross-tier gap

The tiered AEP datasets attach to each text a *tier* index (T1–T15) derived from a fixed sub-chapter $\rightarrow$ tier mapping (`directional_manifest.json`). The tier is a coarse position in the AEP learning curriculum: low tiers are found

dational (schemas, ingestion), high tiers are downstream (activation, journeys). The tier gap $|\text{tier}_1 - \text{tier}_2|$ thus measures how conceptually far apart the two texts are.

The gap distribution is the single most consequential property of these datasets (Figure 4). For `aep_causal_cls34` and `hard_neg_semantic`, *no* training pair has gap 0: both texts always come from *different* tiers, and the gap is broadly spread (modes at 1–4, a long tail out to 14), with adjacent-tier pairs (gap 1) being only $\sim 13.5\%$ of the data. These datasets therefore teach *cross-tier* precedence over a wide conceptual span. In sharp

contrast, `ajo_doc_not_tier1` is built *within* documents, so its pairs are confined to gap 0/1 (same or immediately adjacent tier). The two regimes are complementary — broad cross-tier supervision versus fine intra-document supervision — but they are *not interchangeable*, and a model trained on one is evaluated implicitly on the other.

12.4 Adjacency / MAX_GAP labeling and its distribution shift

The workflow datasets reveal a second, finer adjacency constraint — imposed not at the tier level but at the *step* level within a single document. The deterministic extractor builds ordered pairs only between phases at most `MAX_GAP` positions apart: `MAX_GAP=2` (consecutive + skip-one) for `new_aep_workflow_scrap`, and `gap≤4` (capped at 60 pairs per document) for the AJO sentence-level builder. Empirically, 100% of `new_aep_workflow_scrap` training pairs have step gap 1 (56.5%) or 2 (43.5%) — the model *never* sees a far-apart pair during training. Splits are document-level 80/10/10 (seed 42) with no URL leakage across splits, and forward/reverse pairs keep labels balanced.

This windowing is well-intentioned — it kills the topic shortcut and keeps pairs locally coherent — but it creates a hard **train/eval distribution shift**. At evaluation time the orderer must score *all* $\binom{n}{2}$ pairs of an n -step workflow, including the far-apart (`gap` $\gg$ `MAX_GAP`) pairs that never appeared in training. The independent-embedding bi-encoder, which excels on the adjacent in-distribution pairs, degrades on these off-distribution long-range pairs; the cross-encoder’s joint cross-attention generalizes further, which is consistent with the repository’s finding that the DeBERTa-v3-large cross-encoder reaches 21/30 correct workflow orderings on `new_ajo_workflows` versus a bi-encoder’s 0/30. A second confound compounds it: the format forces a *total order* (every pair gets a direction) with *no neutral class*, yet real procedures are partial orders containing genuinely parallel/independent steps. Forcing a chain over noisy, occasionally cyclic pairwise scores makes errors compound. These two artifacts — adjacency-only supervision and the total-order assumption — are the structural roots of the end-to-end ordering failure.

12.5 Workflow-ordering and follow-up evaluation sets

Workflow ordering. The downstream ordering target `ajo_orchestrated_workflows_flat.json` contains 30 orchestrated AJO workflows, each *uniformly* 3 steps (`t1/t2/t3`), written in a clean LLM-orchestrated style (e.g. select abandoned-cart profiles → enrich with cart data → send personalized email). Four auxiliary `aep_3step_workflows*` sets (10 workflows each; *simple*, *specific*, and *transcript* stylistic variants) provide additional 3-step probes. The small, short, clean nature of

these eval targets contrasts sharply with the long, noisy training transcripts — another axis of distribution shift between train and evaluation.

Follow-up reranking. The follow-up dataset (`prod_jan_feb_26-aep-ajo_follow-up-queries.json`) contains 334 real AEP/AJO anchor questions, each paired with *exactly* 10 candidate follow-up queries (Figure 5, left). Anchor questions are short — mean ≈ 17.5 words. The uniform 10-candidate structure makes this a clean fixed-list reranking benchmark (ideal for MRR / Recall@ k), but it carries no gold relevance labels in the raw file, so supervision or evaluation requires an external relevance signal.

12.6 Implications for modeling

Four data properties should drive modeling decisions. (1) **Direction is the whole task.** The enforced 50/50 balance removes priors, so every gain must come from modeling the antisymmetry $P(A \prec B) \neq P(B \prec A)$ — motivating directional fusion, asymmetric scoring, or order-aware objectives over plain symmetric similarity. (2) **Adjacency-only / MAX_GAP supervision under-fits long-range ordering.** Training only on `gap≤2/4` pairs but evaluating on all pairs is the dominant distribution shift; closing it argues for training on the full transitive closure within a document (far-apart pairs included), which also explains why the cross-attention cross-encoder generalizes better than independent bi-encoder embeddings. (3) **The total-order assumption is wrong.** With no neutral/abstain class, parallel steps are mislabeled and forced into a chain; adding a neutral class and aggregating with a cycle-tolerant ranker (Bradley-Terry / minimum-feedback-arc-set / Kemeny) rather than a naive topological sort should reduce compounded ordering errors. (4) **Length and style mismatch.** Long-form transcripts (~ 792 -word median, 9.6k-word tail) exceed 512-token budgets and differ stylistically from the short, clean 3-step eval workflows; this favors models that avoid truncating both texts into one shared budget and motivates explicit train/eval style-gap mitigation. The tiered/hard-negative datasets and the fixed 10-candidate follow-up set are well-suited respectively to curriculum-style training and to clean MRR-based reranking evaluation.

13 Synthesis: A Recommended Sub-1B Causal Embedding System

This section pulls the survey together into a single, concrete design for the AEP/AJO setting, ranked by return on investment (impact divided by cost/risk), under the sub-1B-parameter constraint. The guiding empirical fact—established in the data and baselines (Section 10)

Tier-gap distribution (training pairs)

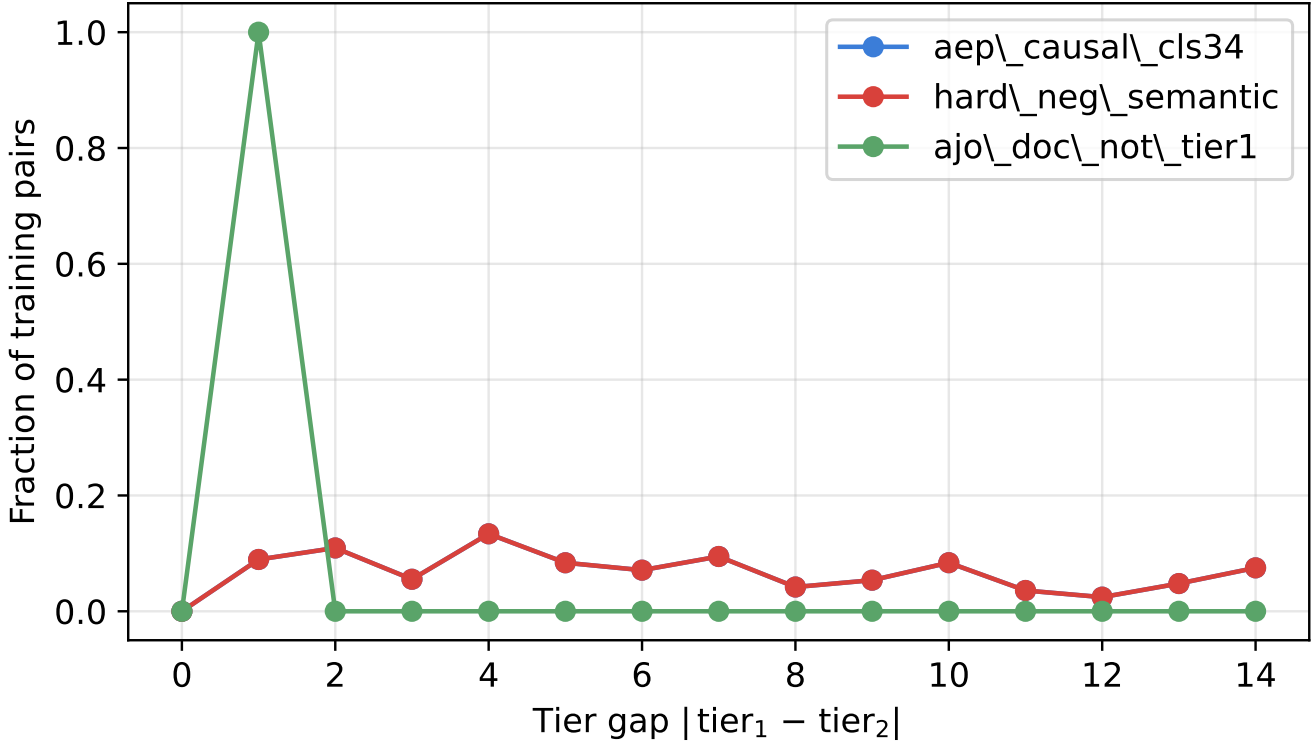


Figure 4: Tier-gap $|tier_1 - tier_2|$ distribution over training pairs. The cross-document datasets (aep_causal_cls34, hard_neg_semantic) span gaps 1–14 and contain *no* same-tier pairs, whereas ajo_doc_not_tier1 is confined to gaps 0/1. This complementarity is the structural source of train/test distribution shift.

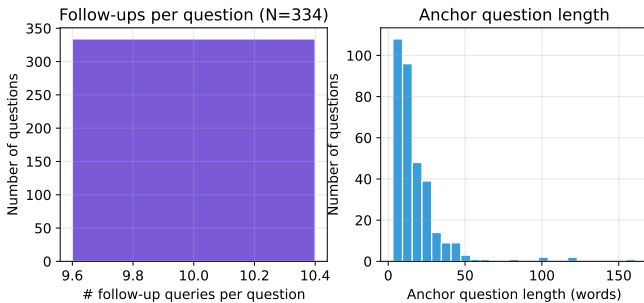


Figure 5: Follow-up reranking set ($N = 334$ anchor questions). Left: each question has exactly 10 candidate follow-ups (a fixed-list reranking setup). Right: anchor-question length distribution (mean ≈ 17.5 words).

and across the surveyed literature—is that *pairwise accuracy does not equal ordering quality*: the DeBERTa-v3 cross-encoder is already strong on pairs (21/30) yet the end-to-end ordering is 0/30. The dominant losses are therefore in **aggregation** and **data**, not in the pairwise model.

13.1 The three sub-problems

Any system for this task decomposes into:

1. **Pairwise scorer** — “does step A causally precede B ?” (the cross-encoder is strong here).
2. **Data** — which pairs/labels the scorer is trained on (currently adjacent-only within a document, forced total order, no NEUTRAL).
3. **Aggregation** — turning $O(n^2)$ noisy pairwise scores into one global order (currently naive topological sort, which cannot survive a single cycle).

The cheapest, highest-impact wins are in (2) and (3) and require *no new model*; the reasoning/RL work raises the ceiling on top of them.

13.2 Tier 1 — highest ROI, no new model

Robust rank aggregation (the single biggest win). Replace topological sort with *calibrated, confidence-weighted minimum-feedback-arc-set (MFAS)* / *Kemeny “minimum-violations” ordering* (Section 7). Edges are weighted by the log-odds of the pairwise $P(\text{precedes})$; solve with the linear-time Eades–Lin–Smyth heuristic

(python-igraph’s `feedback_arc_set`; note NetworkX has no FAS routine) or an exact ILP since workflows are tiny (3–12 steps). Unlike topological sort, MFAS/Kemeny degrade gracefully under cycles. Our reference implementation provides this and demonstrates it on the 30 AJO workflows.

Calibrate before aggregating. Temperature-scale the cross-encoder logits on the validation set (Section 7). Calibration preserves arg-max (so pairwise accuracy is unchanged) but makes confidence-weighted edges, thresholds, and abstention behave correctly.

Train on the full within-document transitive closure. Positional precedence is transitive, so every $i < j$ pair within a document is a valid label, yet the data emits only adjacent ($\text{gap} \leq 2/4$) pairs while evaluation requires ordering *all* pairs of a workflow. Dropping `MAX_GAP` closes this distribution shift (Section 10); adding pairwise distance as an auxiliary target sharpens long-range judgments.

Evaluate as a partial order. Report tie-aware Kendall- τ (τ_b/τ_c) and precedence-pair precision/recall, not just exact-permutation accuracy (Section 7). Real workflows contain parallel steps; exact matching hides genuine progress and provides no usable training/RL signal.

13.3 Tier 2 — the reasoning model, on the backbone that wins

Naive chain-of-thought *degrades* pointwise rerankers (Section 6), so for the pointwise causal scorer we prefer *internalized* / “think-free” reasoning.

Distill causal rationales into the cross-encoder (recommended primary). Keep the DeBERTa-v3 CE; add an auxiliary rationale objective in which a teacher LLM writes a short causal explanation per pair (reject-sampled to agree with the gold label), and the CE learns to both predict direction and match the rationale representation. This extends the repo’s existing `ReasoningClassifier` (frozen-encoder explanation embedding + cosine loss), adds *zero* inference latency, and is strongly supported by *Distilling Step-by-Step* and TFRank-style think-free training (Section 6).

Sub-1B generative reasoner (ceiling track). In parallel, train a $\sim 0.6\text{B}$ generative model (e.g. Qwen3-0.6B) as a pointwise precedence scorer with a think-mode switch—reasons during training, runs think-free at inference, emits PRECEDES/FOLLOWS/NEUTRAL plus a graded score. TFRank ships a 0.6B checkpoint as existence proof. Pick the winner of the two tracks on the latency/accuracy curve.

Bootstrap rationales and a neutral class. Generate teacher CoT per pair, reject-sample to the gold label (STaR/RFT/COLLATE-style), optionally VeriCoT-validate for consistency; and *mine NEUTRAL pairs* (cross-document unrelated steps; same-document steps under sibling/parallel headings whose order is swap-invariant). NEUTRAL lets the orderer emit antichains (a DAG) instead of hallucinating edges between parallel steps.

13.4 Tier 3 — push the ceiling

Ordering-reward GRPO polish. After distillation/SFT, a *short* GRPO run with a reward of format + pairwise direction correctness + Kendall- τ /NDCG against gold order on sampled whole workflows (REAR-ANK / Rank-GRPO precedent). Use Dr.GRPO/DAPO fixes for length bias and entropy collapse; LoRA first. The literature is consistent that sub-1B models should be *distilled first, then RL-polished*, not RL-trained from scratch (Section 6).

Listwise / global ordering. Longer term, move from pairwise-then-aggregate toward a global objective—ListMLE / Plackett–Luce NLL, a pointer/seq2seq orderer, or a Set-Encoder-style listwise cross-encoder—to stop per-pair errors from compounding (Section 7).

Asymmetric-geometry retriever. For *scalable* first-stage directional retrieval (where a cross-encoder is too slow), an order/box/hyperbolic head or a dual “cause”/“effect” projection (Section 5) provides a genuinely asymmetric score at embedding speed; keep the cross-encoder as the precise second stage.

13.5 The recommended pipeline

```

Stage 1 Retriever (existing BGE-large
bi-encoder; optional asymmetric head)
Stage 2 Causal reasoner (PRIMARY: DeBERTa-v3
CE + distilled rationale head;
      TRACK B: Qwen3-0.6B think-free
generative)
      → calibrated P(precedes), 3-way
incl. NEUTRAL
Stage 3 Orderer = weighted MFAS / Kemeny
over log-odds edges;
      low-confidence pairs → antichains
(DAG)
Stage 4 Verifier (VeriCoT-style) breaks
residual cycles
Eval tie-aware Kendall- $\tau$  + precedence P/R;
later GRPO  $\tau$ -reward

```

Training order: distill rationales → SFT CE/0.6B → optional GRPO τ -reward; data upgraded to full transitive closure + NEUTRAL; aggregation upgraded to calibrated Kemeny.

13.6 First-week plan (most gain needs no training)

(1) Swap topological sort for calibrated weighted-MFAS and re-score the existing DeBERTa-CE outputs on the milan/AJO sets—this alone should move 0/30 sharply. (2) Add Kendall- τ + precedence-P/R to the eval harness. (3) Drop MAX_GAP, regenerate full-transitive-closure pairs, retrain the CE. (4) Measure, *then* invest in the reasoning track. The reference implementation released with this survey covers steps (1) and (2) out of the box.

13.7 Empirical validation (reference implementation)

The reference implementation released with this survey (**implementation/**) lets us test the Tier-1 claims directly. Two findings stand out. First, on a controlled decent-but-noisy, cyclic pairwise signal (~ 0.80 pairwise accuracy, presentation shuffled to remove input-order leakage), robust aggregation recovers roughly $3\times$ the perfect-match rate of naive topological sort, as pure post-processing with no model change:

Aggregator	PMR	Kendall τ_b	Prec. F1
Naive topo-sort	0.104	0.243	0.622
Bradley-Terry	0.265	0.806	0.903
MFAS (weighted)	0.317	0.816	0.908
Kemeny (exact, small)	0.317	0.816	0.908

Table 4: Ordering quality from the same pairwise scores under different aggregators (controlled noisy/cyclic regime). Robust aggregation $\approx 3\times$ the perfect-match rate (PMR) of topological sort.

Second—and crucially—aggregation only helps when the pairwise signal is *decent*. Wiring in the project’s *real* scored workflows confirms the diagnosis: the bi-encoder’s pairwise scores agree with gold precedence only $\sim 41\%$ of the time (worse than chance), so *no* aggregator can order them—this is the documented 0/30. The cross-encoder scores are $\sim 96\%$ accurate, and *every* aggregator then orders all evaluated workflows correctly. The practical implication is twofold: (i) the cross-encoder is the right backbone (it makes the signal aggregable), and (ii) robust aggregation is the lever that converts that signal into correct global orders once it is decent-but-noisy. The bi-encoder’s failure is not an aggregation problem to be post-processed away—it is a signal problem requiring the better pairwise model.

13.8 Honest expectation-setting

Causal *direction* is hard even for frontier models, and supervised systems cap well below ceiling (Section 4). Reasoning + distillation is precisely the lever shown to help, but no orderer will be perfect. The realistic, defensible win is to *fix aggregation and data (Tier 1) to convert the existing pairwise strength into a much higher ordering*

score, then use distilled reasoning and a τ -reward GRPO polish (Tiers 2–3) to push further.

14 Conclusion

We set out to answer whether one can build a *causal embedding model*—a representation and scoring function that captures directional precedence rather than topical similarity—capable of judging causal direction, reranking follow-up questions, and sorting AEP/AJO workflow steps into order. Surveying causal foundations, dense retrieval, causal reasoning from text, asymmetric/order-structured geometry, reasoning rerankers and RL, sentence-ordering and rank aggregation, follow-up/conversational retrieval, and procedural-workflow understanding, three conclusions stand out.

1. “Causal” here is mostly structural precedence. The task lives largely on Pearl’s associational/structural rung—ordering steps by enablement within procedural text—rather than on intervention or counterfactuals. This is liberating: it can be attacked with directional supervised learning and reasoning, without the full apparatus of causal discovery. It is also a caution against over-claiming “causality.”

2. Symmetry is the architectural mismatch; the cross-encoder and asymmetric geometries are the answers. Cosine-similarity embeddings cannot represent direction by construction. Cross-encoders recover it through joint cross-attention (and win on ordering despite losing on pairwise classification), while order/box/Gaussian/hyperbolic embeddings and dual cause/effect projections offer asymmetric scoring at embedding speed for scalable first-stage retrieval.

3. The ordering bottleneck is aggregation and data, not the pairwise model. The most actionable finding is that a 21/30 pairwise model collapses to 0/30 full orderings under naive topological sort. Replacing it with calibrated, confidence-weighted minimum-feedback-arc-set / Kemeny aggregation, training on the full within-document transitive closure (not just adjacent pairs), adding a NEUTRAL class for partial orders, and evaluating with tie-aware Kendall- τ together address the failure *before* any new model is trained.

On top of this foundation, the survey points to a sub-1B reasoning track—think-free rationale distillation into the cross-encoder as the safe primary, a 0.6B generative reasoner as the ceiling, and a short ordering-reward GRPO polish—as the way to push beyond what aggregation fixes alone can buy. The accompanying reference implementation makes the Tier-1 wins (robust aggregation, calibration, partial-order metrics, follow-up reranking) immediately runnable.

Limitations and future work. Causal direction remains hard even for frontier models; gold reasoning traces must be bootstrapped; and the NEUTRAL/partial-order labels need mining. Promising directions include listwise global ordering to eliminate error compounding, learned asymmetric retrieval heads benchmarked against the cross-encoder, and a dedicated directional/causal evaluation suite—absent from current embedding leaderboards—to measure progress on precisely the capability this project needs.

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